

Two-dimensional anomalous diffusion of coupled active-passive particles in viscoelastic media

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Anomalous diffusion is well studied in complex environments where viscoelasticity, crowding, and non-equilibrium activity strongly influence particle motion. Several biological systems are not independent; rather, they are coupled to another system, which shows distinct viscoelastic responses and heterogeneous transport behavior. We have developed a theoretical model of an active and a passive particle, coupled via a harmonic interaction, whose motion is restricted to a plane within a viscoelastic medium described by a generalized Langevin framework with a long-tailed power-law memory kernel. We obtained analytical expressions for the mean-squared displacement (MSD) and interparticle separation. We demonstrate that the coupled system shows multiple dynamical crossovers such as subdiffusive, ballistic, and diffusive transport regimes, whose emergence is controlled by the relative viscoelastic exponents and coupling strength. Specifically, we show that long-term dynamics is mainly controlled by the subdiffusive component, where the passive particle can suppress activity and the active particle can dominate and drive enhanced transport. Our results also highlight non-Gaussian displacement statistics at intermediate lag times. These findings provide insight into coupled transport in active matter systems, including tethered biomolecules, molecular motors, cytoskeletal networks and synthetic microswimmers in complex media.

I. INTRODUCTION

The intracellular space of biological cells is a highly concentrated aqueous medium containing biomacromolecules such as proteins and RNA at volume fractions reaching up to 40% [1–5]. This significantly alters the diffusion behavior of biomolecules and plays a crucial role in shaping intracellular transport dynamics. Anomalous diffusion is a well known feature of wide range of natural phenomena which has attracted significant attention over the past decade and across multiple disciplines [6–9]. The main characteristics of anomalous diffusion is that the MSD scales as nonlinear power law, $\langle \delta x^2(t) \rangle \sim t^\beta$ [7, 10, 11]. For subdiffusion the MSD satisfies $0 < \beta < 1$, whereas faster-than-linear scaling, $\beta > 1$, corresponds to superdiffusion. Ultraslow diffusive processes have also been reported where MSD shows logarithmic dependence, $\langle \delta x^2(t) \rangle \sim \log t$ [12, 13]. Many natural systems show characteristics of anomalous diffusion due to a heterogeneous environment, long range correlations, memory effects, and non equilibrium conditions. In these systems, the environment is not purely viscous or elastic, and the stress response exhibits long relaxation, giving rise to viscoelasticity. Examples of viscoelastic materials are lipid bilayers [14, 15], cell components [16–19], polymer networks [20–24], gels [25–27] and crowded suspensions [28–30]. Due to its polymeric nature, the dynamics of DNA chain is predominantly anomalous [31–33]. Also within the classical Rouse polymer model, the dynamics of chain exhibits anomalous behavior with $\beta = 0.5$. Furthermore, the combined effect of hydrodynamic interactions and chain semiflexibil-

ity results in $\beta = 0.75$ [34]. The junctions of the endoplasmic reticulum also exhibit heterogeneous anomalous diffusion with a wide range of anomalous exponents [35]. Anomalous diffusion also plays an important role in environmental and geological systems where transport through porous rocks, soil layers, and groundwater reservoirs deviates from classical diffusion predictions [36, 37]. In addition, anomalous diffusion concepts have been successfully applied in financial modeling, where stock-price fluctuations exhibit long-range correlations and heavy-tailed distributions inconsistent with classical Brownian motion assumptions [38, 39].

Active Brownian motion has received increasing attention among physicists. Unlike passive Brownian particles that are subject to only equilibrium thermal fluctuations, active Brownian particles have a means of self propulsion [40, 41]. Their motion results from the interplay of stochastic fluctuations and intrinsic active motions that pushes the system from thermodynamic equilibrium. Several microscopic living entities employ active brownian motion, one of the best studied examples being the swimming behavior of *Escherichia coli* bacteria [42–45]. In addition, synthetic active particles are viewed as potential candidates for autonomous agents capable of identifying, capturing, and transporting nanoscopic-sized cargo for use in applications such as bioremediation, targeted delivery of drugs, and gene therapy [46–48]. Superdiffusion is typically observed in systems with persistent motion or long range correlations such as turbulent flows [49, 50], animal foraging [51, 52]. An important application of anomalous diffusion arises in active matter such as microswimmers [53, 54], bacteria [55, 56], artificial nanobots moving through complex viscoelastic environment [57, 58]. These artificial microswimmers can achieve propulsion through several different means, including periodic deformations to their shape or the mo-

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tion due to the presence of external fields or gradients [59–62].

Intracellular transport is a particularly rich setting in which both passive and active anomalous diffusion coexist. Passive cargo diffusing in the crowded, viscoelastic cytoplasm typically shows subdiffusive scaling with pronounced non-Gaussian displacement statistics arising from the heterogeneous, dynamic interplay between cytosolic crowding and ATP-driven motion within the cell [63]. In contrast, cargo actively transported along cytoskeletal filaments by molecular motors frequently exhibits superdiffusive, persistent motion, reflecting the directed, motor-driven component of the dynamics [17, 64, 65]. This coexistence of thermally driven subdiffusion and motor-driven superdiffusion within a single crowded, viscoelastic medium is a direct physical realization of the competition between passive viscoelastic memory and active self-propulsion.

The intracellular transport of macromolecules is strongly influenced by specific binding interactions that transiently or permanently couple particles via chemical or physical bonds. A paradigmatic example is a DNA-binding protein in the cell nucleus [66, 67]. Explicit bond-mediated coupling between macromolecules is also observed in membrane-associated systems. The diffusion of peripheral membrane proteins can be governed by their lipid-binding affinities with different binding modes that produce distinct anomalous diffusion signatures [68, 69]. In polymer-protein and cross-linked macromolecular networks, particles are physically tethered via elastic or chemical bonds. In a recent theoretical study, coupled particle system was used to model polymer protein complexes and cross linked cytoskeletal filaments [70]. In highly crowded biological environments, binding mediated coupling can enrich dynamics [71, 72]. Although not always permanent bonds, these interaction potentials effectively act as transient coupling constraints that modulate transport.

Explicit bond-mediated coupling between active and passive units arises in a variety of physical contexts. In cytoskeletal systems, motor proteins are chemically bound to filamentous networks such as actin or microtubules, producing bond-mediated activity transfer central to active gels [73]. Taloni investigated the diffusion of an active particle bound to a generalized elastic system, showing that elastic coupling to passive degrees of freedom produces nontrivial diffusion and anomalous relaxation in the composite system [74]. Dipolar or attractive interactions provide another route to bonding, creating aggregates of active particles coupled to passive or weakly active units; in this vein, Vutukuri *et al.* demonstrated that self-propelled Janus spheres connected via induced bonding interactions exhibit coupled translational and rotational dynamics [75]. More broadly, such coupling also occurs in artificial microswimmers and tethered biomolecular complexes [76–78]. Recent studies of coupled passive-passive and active-passive pairs further show that bonding can generate qualitatively new modes

of motion, with activity transferred through the bond so that the dynamics of each component become strongly interdependent [70, 79–82]. Systems of this kind arise naturally wherever an active unit is elastically bound to a passive viscoelastic partner: molecular motors elastically attached to cargo dragged through the crowded cytoplasm [83, 84], or biohybrid microswimmers bound to soft viscoelastic cores for targeted drug delivery [85]. Active Janus particles embedded in a passive colloidal matrix have likewise been shown to mutually influence each other’s dynamics [86]. These examples motivate the coupled active-passive model developed here.

Under the conditions of anomalous transport, the dynamics is generally described using fractional stochastic frameworks [87–89]. One such theoretical framework used to describe anomalous diffusion in complex environments is the Generalized Langevin equation (GLE), which extends the classical Langevin equation by introducing memory kernels that account for viscoelastic responses of the surrounding medium [64, 90–98]. These kernels often follow power-law behavior, reflecting the presence of multiple interacting spatial and temporal scales within the material. Such modeling approaches are useful for describing particle motion in polymer solutions, lipid membranes, gels, intracellular environments, and crowded suspensions [99–102].

Most of the theoretical and computational investigations in the past have focused on collective behavior or passive responses. The explicit role of active forcing in bonded or mechanically coupled systems has remained underexplored. In the current work, we address this gap by investigating coupled dynamics of active and passive particle connected via a harmonic interaction and moving in two dimensions within a viscoelastic medium. We find that when the active particle’s own MSD scaling exponent is smaller than that of the passive particle, the dynamics of the coupled system is governed primarily by the active particle. Conversely, when the passive particle’s exponent is smaller, its slower dynamics dominates, effectively suppressing the influence of active self-propulsion. These findings show that the interplay between activity and viscoelastic subdiffusion mediated by bonding leads to a nontrivial competition that determines the transport properties of the composite system.

II. MODEL

We investigate the dynamics of coupled active (A) and passive (P) Brownian particles immersed in a viscoelastic medium, as schematically illustrated in Fig. 1. The two particles are coupled by a harmonic force constant k . The state of the active particle is characterized by its position vector $\mathbf{r}_A(t)$ and orientation angle $\phi(t)$, while the state of the passive particle is characterized by its position vector $\mathbf{r}_P(t)$ alone. Let $\hat{\mathbf{n}}(t) = (\cos(\phi(t)), \sin(\phi(t)))$ denote the direction vector of self-propulsion of the active particle, with constant velocity v . The temporal evolution of these

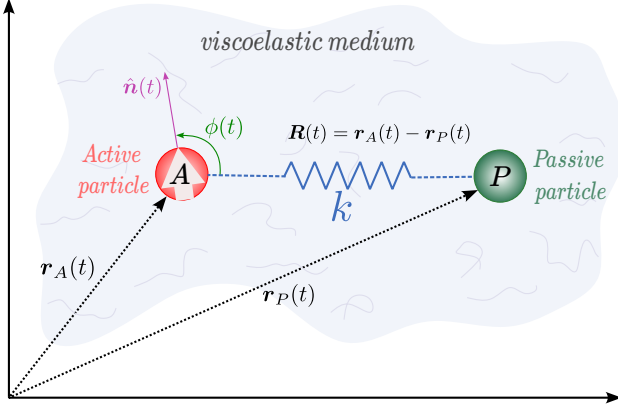


FIG. 1. Schematic representation of the coupled active-passive particle system. The particles are connected by a harmonic spring of stiffness k and move in a plane within a viscoelastic medium; the medium itself need not be two-dimensional. The two-dimensional description is motivated by single-particle-tracking experiments, which often record trajectories as projections onto the focal plane, and by quasi-2D systems such as membrane-bound proteins.

variables is governed by a set of overdamped generalized Langevin equations, which describe the stochastic motion of the system.

$$\int_0^t K_A(t-t') [\dot{\mathbf{r}}_A(t') - v\hat{\mathbf{n}}(t')] dt' = -k[\mathbf{r}_A(t) - \mathbf{r}_P(t)] + \boldsymbol{\xi}_A(t), \quad (1a)$$

$$\int_0^t K_R(t-t') \dot{\phi}(t') dt' = \xi_R(t), \quad (1b)$$

$$\int_0^t K_P(t-t') \dot{\mathbf{r}}_P(t') dt' = -k[\mathbf{r}_P(t) - \mathbf{r}_A(t)] + \boldsymbol{\xi}_P(t). \quad (1c)$$

Here $\boldsymbol{\xi}_A(t)$, $\boldsymbol{\xi}_P(t)$ and $\xi_R(t)$ are zero-mean fractional Gaussian noises that satisfy the fluctuation dissipation theorem (FDT)

$$\begin{aligned} \langle \boldsymbol{\xi}_A(t) \boldsymbol{\xi}_A(t') \rangle &= k_B T \mathbb{I} K_A(|t-t'|), \\ \langle \boldsymbol{\xi}_P(t) \boldsymbol{\xi}_P(t') \rangle &= k_B T \mathbb{I} K_P(|t-t'|), \\ \langle \xi_R(t) \xi_R(t') \rangle &= k_B T K_R(|t-t'|), \end{aligned} \quad (2)$$

where $\mathbb{I}$ is an 2×2 identity matrix, k_B is Boltzmann constant and T is temperature. $K_Q(t)$, where $Q = \{A, P, R\}$, is the memory kernel that governs the viscoelastic response, given by

$$K_Q(t) = \frac{\eta_Q t^{-\alpha_Q}}{\Gamma(1-\alpha_Q)}. \quad (3)$$

Here $\alpha_Q \in (0, 1)$, $\Gamma(z)$ is the Gamma function, $\eta_Q = \gamma_Q \Gamma(3-\alpha_Q)$, and γ_Q is the generalized friction coefficient. The power-law kernel $K_Q(t)$ arises generically from

Mori-Zwanzig (MZ) projection when the surrounding bath possesses a continuous, power-law-distributed spectrum of relaxation times, corresponding to a frequency-dependent spectral density $J(\omega) \propto \omega^{1-\alpha_Q}$ [87, 89, 103–106]. This construction generalizes the Ohmic (Markovian) case: in the limit of Gaussian white uncorrelated noise ($\alpha_Q = 1$), $K_Q(t) = 2\gamma_Q \delta(t)$.

This choice of kernel is also the physically appropriate one for the systems motivating this work. A single-relaxation-time Maxwell kernel, which decays exponentially and yields no anomalous scaling, is suited to simple viscoelastic fluids with one dominant relaxation mode. Crowded cytoplasm, cross-linked polymer and cytoskeletal networks, and gels, by contrast, are characterized by power-law, rather than Debye/Maxwell, frequency-dependent viscoelastic moduli over several decades [107, 108], with directly measured MSD scaling exponents $\beta \approx 0.4$ reported for chromosomal loci in living cells [32]. The soft glassy rheology framework [109] formalizes why such broad relaxation-time distributions, rather than single-mode Maxwell behavior, are the generic case in disordered, crowded soft matter. In this sense, α_Q quantifies the breadth and self-similarity of the relaxation-time spectrum of the surrounding medium, for example, a fractal or entangled polymer network, a crosslinked cytoskeleton, or a crowded cytoplasm [27, 32, 34, 64, 90, 92, 110].

Equation (2) constrains only the thermal bath noises $\boldsymbol{\xi}_A(t)$, $\boldsymbol{\xi}_P(t)$, $\xi_R(t)$, describing an equilibrium viscoelastic medium unaffected by the particle's activity. The self-propulsion force $v\hat{\mathbf{n}}(t)$ in Eq. (1a) is a non-thermal driving term added on top of this equilibrium bath, following the standard active-particle construction [40, 111, 112]; the system is thereby driven out of equilibrium even though the bath itself satisfies FDT.

All quantities carry physical dimensions in the standard way, summarized in Table I. The one quantity requiring comment is the generalized friction coefficient γ_Q : unlike an ordinary friction coefficient, it does not carry units of mass/time, but rather $\text{kg} \cdot \text{s}^{\alpha_Q-2}$, reducing to the familiar kg/s only in the Markovian limit $\alpha_Q \rightarrow 1$.

We restrict the motion of the dimer to a plane, while the surrounding medium itself need not be two-dimensional. This setting is motivated by single-particle-tracking experiments, which often record trajectories as projections onto the microscope focal plane [32], and by genuinely quasi-2D systems such as membrane-bound or cortex-associated proteins [69, 113].

The Laplace transform of any function $f(t)$ is defined as $\tilde{f}(s) = \int_0^\infty e^{-st} f(t) dt$, and is indicated by the tilde symbol throughout the paper. Then, Eq. (1) can be written in Laplace space as follows

TABLE I. Physical units of the main quantities used in the text.

Quantity	Units
$\mathbf{r}_A, \mathbf{r}_P, \mathbf{R}$	length
$t, \tau_A, \tau_P, \tau_v, \tau_l$	time
$k_B T$	energy
k, k_c	force/length
v	length/time
α_Q	dimensionless
$\gamma_A, \gamma_P, \eta_A, \eta_P$	$\text{kg} \cdot \text{s}^{\alpha_Q-2}; \quad Q \in \{A, P\}$
γ_R, η_R	$\text{kg} \cdot \text{m}^2 \cdot \text{s}^{\alpha_R-2}$
D_R	$\text{rad}^2/\text{s}^{\alpha_R}$

$$\tilde{K}_A(s) [\tilde{\mathbf{r}}_A(s) - v\tilde{\mathbf{n}}(s)] = -k[\tilde{\mathbf{r}}_A(s) - \tilde{\mathbf{r}}_P(s)] + \tilde{\boldsymbol{\xi}}_A(s), \quad (4a)$$

$$\tilde{K}_R(s) \tilde{\dot{\phi}}(s) = \tilde{\xi}_R(s), \quad (4b)$$

$$\tilde{K}_P(s) \tilde{\mathbf{r}}_P(s) = -k[\tilde{\mathbf{r}}_P(s) - \tilde{\mathbf{r}}_A(s)] + \tilde{\boldsymbol{\xi}}_P(s). \quad (4c)$$

A. MSD of active and passive particle

We define the interparticle separation as

$$\mathbf{R}(t) = \mathbf{r}_A(t) - \mathbf{r}_P(t).$$

Using the Laplace transform of the position vectors in terms of their time derivatives, $\tilde{\mathbf{r}}_A(s) = (\tilde{\mathbf{r}}_A(s) + \mathbf{r}_A(0))/s$ and $\tilde{\mathbf{r}}_P(s) = (\tilde{\mathbf{r}}_P(s) + \mathbf{r}_P(0))/s$, the Laplace-space separation is

$$\tilde{\mathbf{R}}(s) = \tilde{\mathbf{r}}_A(s) - \tilde{\mathbf{r}}_P(s) = \frac{\tilde{\mathbf{r}}_A(s) - \tilde{\mathbf{r}}_P(s) + \mathbf{R}(0)}{s}. \quad (5)$$

Using Eq. (5) in Eq. (4a) and 4c, and after rearranging, we can write

$$\begin{aligned} \tilde{\mathbf{r}}_A(s) = & \frac{k[\tilde{\mathbf{r}}_P(s) - \mathbf{R}(0)]}{s\tilde{K}_A(s) + k} + v\tilde{\mathbf{n}}(s) \frac{s\tilde{K}_A(s)}{s\tilde{K}_A(s) + k} \\ & + \frac{s\tilde{\boldsymbol{\xi}}_A(s)}{s\tilde{K}_A(s) + k}, \end{aligned} \quad (6)$$

$$\tilde{\mathbf{r}}_P(s) = \frac{k[\tilde{\mathbf{r}}_A(s) + \mathbf{R}(0)]}{s\tilde{K}_P(s) + k} + \frac{s\tilde{\boldsymbol{\xi}}_P(s)}{s\tilde{K}_P(s) + k}. \quad (7)$$

Using Eqs. (6) and (7), we can obtain the expression of $\tilde{\mathbf{r}}_A(s)$ and $\tilde{\mathbf{r}}_P(s)$ in Laplace space. The resulting ex-

pressions are given below.

$$\begin{aligned} \tilde{\mathbf{r}}_A(s) = & \frac{\mathbf{r}_A(0)}{s} - \frac{\tilde{\Psi}_P(s)\mathbf{R}(0)}{s(\tilde{K}_A(s) + \tilde{\Psi}_P(s))} + \frac{v\tilde{\mathbf{n}}(s)\tilde{K}_A(s)}{s(\tilde{K}_A(s) + \tilde{\Psi}_P(s))} \\ & + \frac{\frac{\tilde{\Psi}_P(s)}{\tilde{K}_P(s)}\tilde{\boldsymbol{\xi}}_P(s) + \tilde{\boldsymbol{\xi}}_A(s)}{s(\tilde{K}_A(s) + \tilde{\Psi}_P(s))}, \end{aligned} \quad (8)$$

$$\begin{aligned} \tilde{\mathbf{r}}_P(s) = & \frac{\mathbf{r}_P(0)}{s} + \frac{\tilde{\Psi}_A(s)\mathbf{R}(0)}{s(\tilde{K}_P(s) + \tilde{\Psi}_A(s))} + \frac{v\tilde{\mathbf{n}}(s)\tilde{\Psi}_A(s)}{s(\tilde{K}_P(s) + \tilde{\Psi}_A(s))} \\ & + \frac{\frac{\tilde{\Psi}_A(s)}{\tilde{K}_A(s)}\tilde{\boldsymbol{\xi}}_A(s) + \tilde{\boldsymbol{\xi}}_P(s)}{s(\tilde{K}_P(s) + \tilde{\Psi}_A(s))}, \end{aligned} \quad (9)$$

where we defined

$$\tilde{\Psi}_A(s) = \frac{k\tilde{K}_A(s)}{s\tilde{K}_A(s) + k}, \quad \tilde{\Psi}_P(s) = \frac{k\tilde{K}_P(s)}{s\tilde{K}_P(s) + k}. \quad (10)$$

The functions $\tilde{\Psi}_A(s)$ and $\tilde{\Psi}_P(s)$ characterize the mutual coupling between active and passive particles, capturing their reciprocal influence on each other. In the absence of coupling, $k = 0$, both functions vanish and the dynamics reduces to those of two uncoupled particles.

$$\begin{aligned} \tilde{\mathbf{r}}_A(s) = & \frac{\mathbf{r}_A(0)}{s} + \frac{v\tilde{\mathbf{n}}(s)}{s} + \frac{\tilde{\boldsymbol{\xi}}_A(s)}{s\tilde{K}_A(s)}, \\ \tilde{\mathbf{r}}_P(s) = & \frac{\mathbf{r}_P(0)}{s} + \frac{\tilde{\boldsymbol{\xi}}_P(s)}{s\tilde{K}_P(s)}. \end{aligned} \quad (11)$$

To improve the clarity of the mathematical procedure, we define some auxiliary functions

$$\tilde{\Theta}_A(s) = \frac{1}{s(\tilde{K}_A(s) + \tilde{\Psi}_P(s))}, \quad \tilde{\Delta}_A(s) = \tilde{K}_A(s)\tilde{\Theta}_A(s), \quad (12)$$

$$\tilde{\Theta}_P(s) = \frac{1}{s(\tilde{K}_P(s) + \tilde{\Psi}_A(s))}, \quad \tilde{\Delta}_P(s) = \tilde{\Psi}_A(s)\tilde{\Theta}_P(s). \quad (13)$$

Using Eqs. (8), (9), and (2) in Laplace space, together with the equilibrium interparticle separation $\langle \mathbf{R}^2(0) \rangle = 2k_B T/k$, and taking the inverse double Laplace transform (see Appendix A for details), the mean-squared displacement of the active particle,

$$\langle \delta \mathbf{r}_A^2(t) \rangle = \langle (\mathbf{r}_A(t) - \mathbf{r}_A(0))^2 \rangle,$$

can be written as

$$\begin{aligned} \langle \delta \mathbf{r}_A^2(t) \rangle = & 4k_B T \int_0^t dt' \Theta_A(t') \\ & + v^2 \int_0^t dt' \int_0^t d\tau' \Delta_A(t') \Delta_A(\tau') C(|t' - \tau'|). \end{aligned} \quad (14)$$

where $C(|t - \tau|) = \langle \hat{\mathbf{n}}(t) \cdot \hat{\mathbf{n}}(\tau) \rangle$. The first term on the right-hand side corresponds to the thermal contribution to the MSD, and the second term corresponds to the contribution of the active force. Similarly, the MSD of the passive particle, $\langle \delta \mathbf{r}_P^2(t) \rangle = \langle (\mathbf{r}_P(t) - \mathbf{r}_P(0))^2 \rangle$, can be written as

$$\langle \delta \mathbf{r}_P^2(t) \rangle = 4k_B T \int_0^t dt' \Theta_P(t') + v^2 \int_0^t dt' \int_0^t d\tau' \Delta_P(t') \Delta_P(\tau') C(|t' - \tau'|). \quad (15)$$

B. MSD of interparticle distance

We defined $\mathbf{R}(t) = \mathbf{r}_A(t) - \mathbf{r}_P(t)$ as the intermolecular distance between the active and passive particles. Dividing Eq. (4c) and Eq. (4a) by their respective kernels in Laplace space, subtracting, and rearranging, we obtain the following

$$\tilde{K}_{eff}(s) \tilde{\mathbf{R}}(s) = v \tilde{K}_{eff}(s) \tilde{\mathbf{n}}(s) - k \tilde{\mathbf{R}}(s) + \tilde{\boldsymbol{\xi}}_{eff}(s), \quad (16)$$

where

$$\tilde{K}_{eff}(s) = \frac{\tilde{K}_A(s) \tilde{K}_P(s)}{\tilde{K}_A(s) + \tilde{K}_P(s)}, \quad (17)$$

and

$$\tilde{\boldsymbol{\xi}}_{eff}(s) = \left(\frac{\tilde{\boldsymbol{\xi}}_A(s)}{\tilde{K}_A(s)} - \frac{\tilde{\boldsymbol{\xi}}_P(s)}{\tilde{K}_P(s)} \right) \tilde{K}_{eff}(s). \quad (18)$$

From Eq. (16), we can write

$$\int_0^t K_{eff}(t - t') [\dot{\mathbf{R}}(t') - v \hat{\mathbf{n}}(t')] dt' = -k \mathbf{R}(t) + \boldsymbol{\xi}_{eff}(t). \quad (19)$$

where the effective noise $\boldsymbol{\xi}_{eff}(t)$ satisfies the fluctuation-dissipation theorem,

$$\langle \boldsymbol{\xi}_{eff}(t) \boldsymbol{\xi}_{eff}(t') \rangle = k_B T \mathbb{I} K_{eff}(|t - t'|). \quad (20)$$

From Eq. (16) we obtain the mean squared distance between active and passive particles, $\langle \delta \mathbf{R}^2(t) \rangle = \langle (\mathbf{R}(t) - \mathbf{R}(0))^2 \rangle$, as (see Appendix B for detailed calculations)

$$\langle \delta \mathbf{R}^2(t) \rangle = \frac{4k_B T}{k} [1 - \Phi(t)] + v^2 \int_0^t dt' \int_0^t d\tau' C(|t' - \tau'|) \Phi(t') \Phi(\tau'), \quad (21)$$

where $\Phi(t)$ is the inverse Laplace transform of $\tilde{\Phi}(s)$, written as

$$\tilde{\Phi}(s) = \frac{\tilde{K}_{eff}(s)}{s \tilde{K}_{eff}(s) + k}. \quad (22)$$

C. Correlation between active particle orientations

Now we will calculate the correlation function $C(|t - \tau|) = \langle \hat{\mathbf{n}}(t) \cdot \hat{\mathbf{n}}(\tau) \rangle$, using the assumption that noise is Gaussian. For a Gaussian random variable X , with $\langle X \rangle = 0$ and $\langle X^2 \rangle = \sigma^2$, the characteristic function is written as

$$\langle e^{\pm iX} \rangle = e^{-\frac{\sigma^2}{2}}. \quad (23)$$

Therefore,

$$\langle \cos(X) \rangle = \frac{1}{2} (\langle e^{+iX} \rangle + \langle e^{-iX} \rangle) = e^{-\frac{\sigma^2}{2}}. \quad (24)$$

From the definition of $\hat{\mathbf{n}}(t)$ and using trigonometric identity

$$\langle \hat{\mathbf{n}}(t) \cdot \hat{\mathbf{n}}(\tau) \rangle = \langle \cos(\phi(t) - \phi(\tau)) \rangle = \langle \cos(\Delta\phi) \rangle. \quad (25)$$

After solving Eq. (4b) we can calculate the mean squared displacement of angle ϕ (see Appendix C for detailed calculation), given as

$$\langle (\phi(t) - \phi(\tau))^2 \rangle = \frac{2k_B T}{\eta_R \Gamma(1 + \alpha_R)} |t - \tau|^{\alpha_R}. \quad (26)$$

Then

$$C(|t - \tau|) = \exp\left(-\frac{k_B T}{\eta_R \Gamma(1 + \alpha_R)} |t - \tau|^{\alpha_R}\right). \quad (27)$$

The inverse Laplace transforms of $\Delta_A(t)$, $\Delta_P(t)$, $\Theta_A(t)$, $\Theta_P(t)$, and $\Phi(t)$ were computed numerically using the de Hoog algorithm [114], as implemented in the `invertlaplace` routine of the `mpmath` Python library [115], with working precision set to 25 decimal digits. Laplace-space functions were evaluated on a logarithmically spaced time grid spanning $t \in [10^{-10}, 10^6]$ with 1000 points. To confirm the robustness of this choice, we cross-validated the de Hoog results against two independent inversion algorithms, fixed Talbot and Gaver–Stehfest. All three methods agree closely. The double time integrals appearing in the active contributions to the MSD were evaluated by direct quadrature (trapezoidal rule) over the overlapping domain of the two time arguments, with the inverse-Laplace-transformed response functions $\Delta_A(t)$, $\Delta_P(t)$ interpolated onto the integration grid. The short-time contribution to the thermal integral, $\int_0^t dt' \Theta_A(t')$ for $t \rightarrow 0$, was evaluated using the known analytic short-time behavior $\Theta_A(t) \simeq t^{\alpha_A - 1} / (\eta_A \Gamma(\alpha_A))$ rather than direct numerical integration through the (integrable) singularity at $t' = 0$. All numerical results were validated against the exact analytic limits available in closed form, the uncoupled ($k = 0$) result.

III. RESULTS AND DISCUSSION

A. MSD

The function $\Theta_{A/P}(t)$ gives the contribution of thermal fluctuations in MSD of active and passive parti-

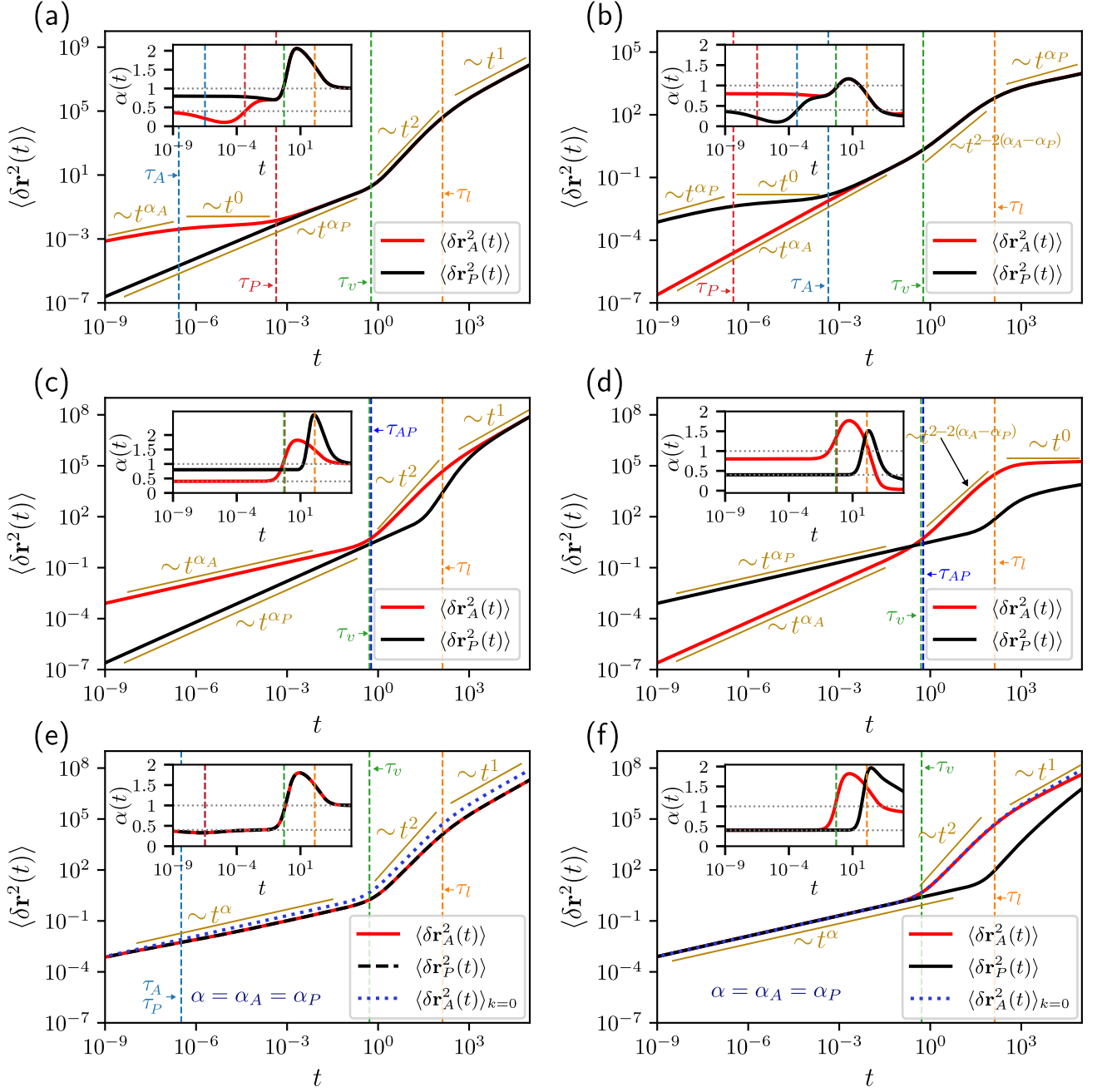


FIG. 2. MSD of active and passive particles for different cases. (a) $\alpha_A = 0.4, \alpha_P = 0.8, k = 500$, (b) $\alpha_A = 0.8, \alpha_P = 0.4, k = 500$ (c) $\alpha_A = 0.4, \alpha_P = 0.8, k = 0.01$, (d) $\alpha_A = 0.8, \alpha_P = 0.4, k = 0.01$ (e) $\alpha_A = 0.4, \alpha_P = 0.4, k = 500$, (f) $\alpha_A = 0.4, \alpha_P = 0.4, k = 0.01$ The inset plot shows $\alpha(t)$ for corresponding color. Vertical dashed lines indicate the characteristic timescales. The fixed parameters used in the figures are $k_B T = 1, v = 3, \gamma_A = 1, \gamma_P = 1, \alpha_R = 0.5, \gamma_R = 4$.

cles. For uncoupled system, $k = 0$, therefore, $\Theta_A(t) = t^{\alpha_A - 1}/(\eta_A \Gamma(\alpha_A))$ and $\Delta_A(t) = 1$. Consequently, the

MSD for active particle reduces to (see Appendix E)

$$\begin{aligned} \langle \delta \mathbf{r}_A^2(t) \rangle_{k=0} = & \frac{4k_B T t^{\alpha_A}}{\eta_A \Gamma(1 + \alpha_A)} + \frac{2v^2}{\alpha_R} \left[t D_R^{-1/\alpha_R} \gamma\left(\frac{1}{\alpha_R}, D_R t^{\alpha_R}\right) \right. \\ & \left. - D_R^{-2/\alpha_R} \gamma\left(\frac{2}{\alpha_R}, D_R t^{\alpha_R}\right) \right], \end{aligned} \quad (28)$$

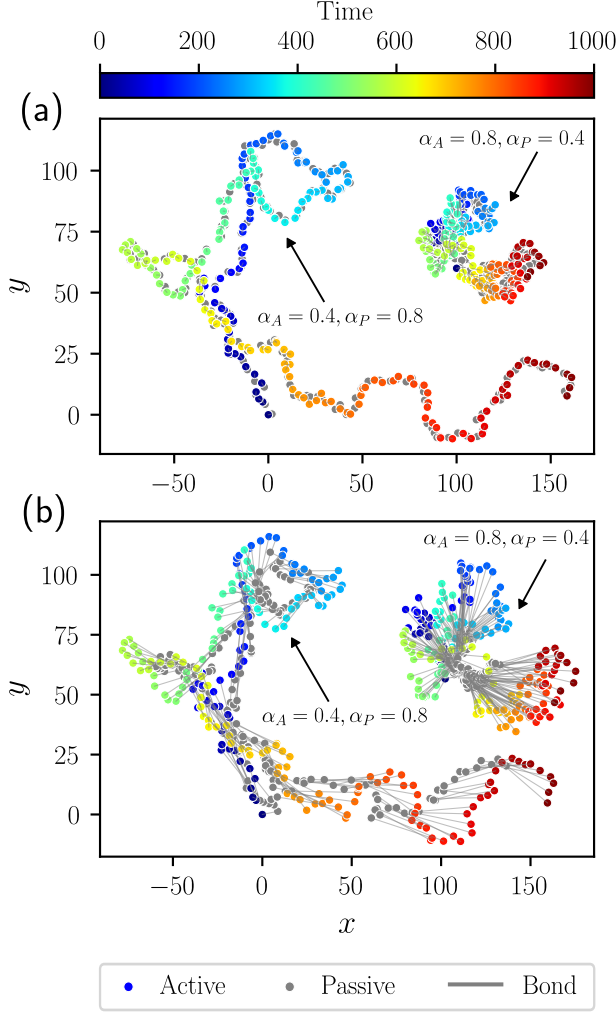


FIG. 3. Simulated trajectories of coupled system for two cases (a) strong interaction ($k \gg k_c$) and (b) weak interaction ($k \ll k_c$). Color of active particle shows the progression of time according to the colorbar. Corresponding cases have parameters from Fig. 2.

where

$$D_R = \frac{k_B T}{\eta_R \Gamma(1 + \alpha_R)}, \quad (29)$$

and $\gamma(b, z)$ is lower incomplete Gamma function. This result has also been derived in ref. [98]. Similarly, for passive MSD

$$\langle \delta \mathbf{r}_P^2(t) \rangle_{k=0} = \frac{4k_B T}{\eta_P \Gamma(1 + \alpha_P)} t^{\alpha_P}. \quad (30)$$

To separate different regimes of motion, we define characteristic timescales for timescale of active and passive particles

$$\tau_A = \left(\frac{\eta_A \Gamma(1 + \alpha_A)}{k} \right)^{1/\alpha_A}, \quad \tau_P = \left(\frac{\eta_P \Gamma(1 + \alpha_P)}{k} \right)^{1/\alpha_P}. \quad (31)$$

These indicate the time it takes by the free active and passive particles to reach the mean confinement distance $4k_B T/k$ (for the 2D system) in the harmonic potential in equilibrium. Depending on the values of k , the harmonic constraint can be weak or strong. To quantify the strength of this constraint, we define k_c , which represents the spring constant when $\tau_A = \tau_P$.

$$k_c = \left(\frac{(\eta_A \Gamma(1 + \alpha_A))^{\alpha_P}}{(\eta_P \Gamma(1 + \alpha_P))^{\alpha_A}} \right)^{1/(\alpha_P - \alpha_A)}, \quad (32)$$

The exponent $1/(\alpha_P - \alpha_A)$ diverges as $\alpha_A \rightarrow \alpha_P$ which reflects the fact that when both particles share the same viscoelastic exponent, the ratio τ_A/τ_P becomes independent of k , so no finite coupling strength causes the two confinement timescales to cross. At $k = k_c$, the relaxation times of each particle in a harmonic bond is equal. Therefore, $k < k_c$ and $k > k_c$ can be considered as weak and strong couplings, respectively [70]. When the MSD of two freely diffusing particles becomes equal, we define the time at which their MSDs intersect as:

$$\tau_{AP} = \left(\frac{\eta_P \Gamma(1 + \alpha_P)}{\eta_A \Gamma(1 + \alpha_A)} \right)^{1/(\alpha_P - \alpha_A)}. \quad (33)$$

This determines the timescales at which the dynamics of the two anomalously diffusing particles become comparable. To characterize the instantaneous dynamical regime of the coupled active-passive system, we introduce the time-dependent scaling exponent $\alpha(t)$, defined as the logarithmic derivative of the MSD with respect to time,

$$\alpha(t) = \frac{d \log \langle \delta \mathbf{r}^2(t) \rangle}{d \log t}. \quad (34)$$

This quantity is a direct measure of the local transport behavior at each time scale that can allow us to resolve multiple crossover regimes arising from the combined effects of viscoelastic memory, harmonic coupling, and active self-propulsion.

We first analyze two different cases corresponding to the relative memory exponents of the active and passive particles, $\alpha_A < \alpha_P$ and $\alpha_A > \alpha_P$ in the strong interaction limit ($k \gg k_c$).

1. Case: $k \gg k_c$ and $\alpha_A < \alpha_P$

We begin with the case $\alpha_A < \alpha_P$ with strong harmonic interaction. As illustrated in Fig. 2a, MSD of the active particle exhibits multiple dynamical regimes at different time scales. At very short times, both the interaction with the passive particle and the effect of active self-propulsion are negligible. Therefore, the active particle displays free subdiffusive motion governed solely by the viscoelastic environment. In this regime, the MSD scales as

$$\langle \delta \mathbf{r}_A^2(t) \rangle \simeq \frac{4k_B T}{\eta_A \Gamma(1 + \alpha_A)} t^{\alpha_A}. \quad (35)$$

This behavior persists up to the characteristic time scale τ_A , beyond which the effect of confinement due to harmonic coupling becomes significant. As a result, the growth of the MSD slows down and the local scaling exponent $\alpha(t)$ (see inset) decreases. In the limit of very strong confinement, the MSD approaches a plateau value given approximately by

$$\langle \delta \mathbf{r}_A^2(t) \rangle \simeq \frac{4k_B T}{k}. \quad (36)$$

At the characteristic time scale τ_P , the passive particle begins to experience the influence of the harmonic interaction. In this intermediate regime, the dynamics of the active particle becomes effectively governed by the viscoelastic response of the passive particle, and the MSD follows the scaling

$$\langle \delta \mathbf{r}_A^2(t) \rangle \simeq \frac{4k_B T}{\eta_P \Gamma(1 + \alpha_P)} t^{\alpha_P}. \quad (37)$$

This behavior continues until the time scale

$$\tau_v = (4k_B T / (v^2 \eta_A \Gamma(1 + \alpha_A)))^{1/(2 - \alpha_A)}, \quad (38)$$

until which the active self propulsion force is insignificant. This time scale is defined where the free subdiffusion term is balanced by a negative term in Eq. (28). Physically, this balances the thermal term against the short-time ballistic limit of the active term. Beyond this point, the contribution of the active self-propulsion force on the active particle becomes dominant, and the MSD enters a ballistic regime, characterized by $\langle \delta \mathbf{r}_A^2(t) \rangle \propto t^2$. The quadratic growth of the MSD reflects the persistence of the self propulsion direction where the active particle effectively performs directed motion.

At longer times, the persistence of the propulsion direction is gradually lost due to rotational diffusion. The relevant crossover occurs at persistence time

$$\tau_l = D_R^{-1/\alpha_R} \Gamma(2/\alpha_R) / \Gamma(1/\alpha_R), \quad (39)$$

where frequent random reorientation of the propulsion direction suppresses the persistent motion. As a consequence, ballistic growth can no longer be sustained, and dynamics crosses over to an effective normal diffusive regime with $\langle \delta \mathbf{r}_A^2(t) \rangle \propto t$. At τ_l , the positive incomplete gamma function term starts to dominate over the negative one Eq. (28), i.e., $\tau_l \gamma(1/\alpha_R, D_R \tau_l^{\alpha_R}) \approx D_R^{-1/\alpha_R} \gamma(2/\alpha_R, D_R \tau_l^{\alpha_R})$.

Until $t < \tau_P$ the passive particle imposes a confining effect on the active particle. However, beyond this timescale the dynamics become dominated by the self propulsion force of active particle. As a consequence, the motions of the two particles become strongly correlated, resulting in identical dynamical behavior for the active and passive particles. This regime is reflected in the simulated trajectory, Fig. 3a, where the motions of the active and passive particles exhibit a strong correlation. Moreover, the presence of an active driving force leads to more persistent dynamics.

2. Case: $k \gg k_c$ and $\alpha_A > \alpha_P$

We now consider the second case, $\alpha_A > \alpha_P$ with strong interaction, see Fig. 2b. In this regime, the MSD of the active particle remains comparatively low, creating a confining effect on the passive particle. The dynamics of the active and passive particles are similar, with their roles reversed, until the active force begins to influence the system for $t < \tau_v$.

The influence of the active propulsion force shows different characteristics. Due to the sluggish motion of the passive particle, the influence of active propulsion is suppressed by coupling. After time scale $t > \tau_v$ the contribution of the active self-propulsion force becomes significant and enhances the growth of the MSD. However, the corresponding local scaling exponent does not reach the ballistic limit. This can be understood directly from Eq. (14). In the regime $t > \tau_v$, the active contribution to $\langle \delta \mathbf{r}_A^2(t) \rangle$ is governed by the double integral of $\Delta_A(t') \Delta_A(\tau') C(|t' - \tau'|)$. When $\alpha_A > \alpha_P$, the response function $\Delta_A(t)$ does not saturate at unity but instead decays as $\Delta_A(t) \simeq (\eta_A / \eta_P) t^{-(\alpha_A - \alpha_P)} / \Gamma(1 + \alpha_P - \alpha_A)$ at long times (see Fig. 4, green dotted line, also Table II), reflecting the persistent confining influence of the more slowly relaxing passive particle. Since the orientational correlation $C(|t' - \tau'|)$ remains close to unity over the timescale of interest (i.e., before the persistence time τ_l), the double integral scales as $t^2 \times \Delta_A(t)^2 \sim t^{2 - 2(\alpha_A - \alpha_P)}$, rather than the free ballistic result t^2 obtained when $\Delta_A(t) \rightarrow 1$. The local scaling exponent is therefore set by $2 - 2(\alpha_A - \alpha_P)$, consistent with Fig. 2(b), and reflects the fact that the active driving is only partially transmitted to the active particle's own MSD once it is dynamically enslaved to the slower passive partner.

After τ_l , the MSD is dominated by a passive particle leading to a subdiffusive regime characterized by $\langle \delta \mathbf{r}_A^2(t) \rangle \propto t^{\alpha_P}$. This result is particularly notable, as it illustrates that the longer time dynamics of the active particle become effectively governed by the viscoelastic properties of the passive particle. For a visual representation of this case, refer to the trajectory shown in Fig. 3a, where persistent motion is absent.

3. Case: $k \ll k_c$ and $\alpha_A < \alpha_P$

We now consider the weak interaction regime, $k \ll k_c$ (Figs. 2c and d). In contrast to the strong-coupling case, the plateau observed in the MSD of both active and passive particles is absent. This lack of saturation reflects the inability of either particle to generate sufficiently strong harmonic confinement to restrict the motion of the other.

At short times, both particles exhibit independent subdiffusive behavior and converge to the same MSD at $t \approx \tau_{AP}$, which represents the characteristic timescale at which two freely diffusing particles attain comparable displacements. At longer times, $t > \tau_v$, the dynamics of

the active particle is dominated by self-propulsion. In the case $\alpha_A < \alpha_P$ (Fig. 2c), this leads to a ballistic regime, $\langle \delta \mathbf{r}_A^2(t) \rangle \propto t^2$. The influence of the active force on the passive particle is correspondingly delayed, but at sufficiently long times, both particles exhibit normal diffusive motion and move in a correlated manner.

4. Case: $k \ll k_c$ and $\alpha_A > \alpha_P$

In contrast, when $\alpha_A > \alpha_P$, the motion of the active particle is suppressed by passive particle, see Fig. 2d. As a result, the active particle experiences an effective confinement, and its MSD approaches a plateau.

These different regimes are illustrated in the simulated trajectories (Fig. 3b). In the former case, the active particle can drag the passive particle along and results in correlated motion over longer distances. In the latter case, although the active particle tends to explore larger displacements, it remains effectively anchored to the passive particle, which exhibits limited mobility. This leads to an apparent confinement that is reflected in the saturation of the MSD.

These dynamical scenarios can also be understood through the response function $\Delta_A(t)$, which quantifies the contribution of the active force to the active particle's motion (Fig. 4). At short times, $\Delta_A(t \rightarrow 0) \approx 1$, indicating maximal active contribution, though its effect on the MSD is masked by dominant subdiffusion at these scales. As t increases, $\Delta_A(t)$ decreases due to the confining influence of the harmonic coupling. For $\alpha_A < \alpha_P$ under strong coupling, $\Delta_A(t)$ falls before rising again and approaching unity at long times (red curve), signaling the eventual restoration of the active contribution; under weak coupling, $\Delta_A(t) \approx 1$ throughout. For $\alpha_A > \alpha_P$, by contrast, $\Delta_A(t)$ decreases monotonically and vanishes at long times, reflecting the persistent suppression of the active force by the more sluggish passive particle.

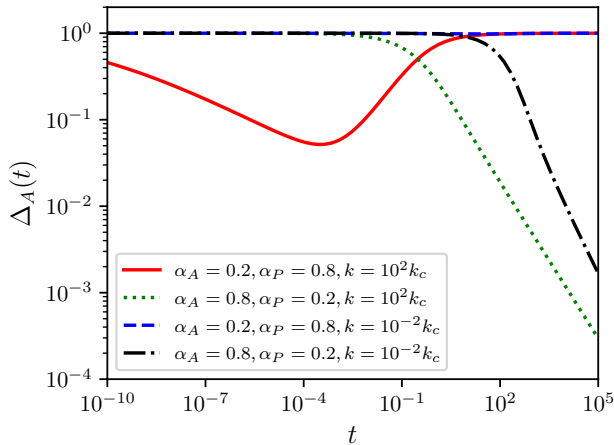


FIG. 4. $\Delta_A(t)$ as a function of t for different cases where $\gamma_A = \gamma_P = 1$.

5. Case: $\alpha_A = \alpha_P$

It is also instructive to analyze the case in which both particles share the same viscoelastic exponent, $\alpha_A = \alpha_P$, corresponding to a medium that acts identically on both particles. This is the physically symmetric limit in which the coupling constant k_c is no longer well defined in Eq. (32).

In the strong-coupling limit (large k), the two particles move rigidly together and their MSDs are close, as shown in Fig. 2(e). The resulting MSD lies close to that of a free active particle ($k = 0$), but with reduced amplitude, since the passive particle must now be dragged along by the active driving force, effectively increasing the drag experienced by the coupled pair.

In the weak-coupling limit (small k), shown in Fig. 2(f), the active particle's MSD is essentially unaffected by the coupling and coincides with that of a free active particle. The passive particle, in contrast, initially follows its own free subdiffusive scaling, $\langle \delta \mathbf{r}_P^2(t) \rangle \sim t^{\alpha_P}$, since the harmonic coupling is too weak to transmit the active driving force at short times. At longer times, however, the weak coupling progressively transmits the active particle's persistent motion to the passive particle, and the two MSDs gradually converge.

B. Mean Squared Separation

We now analyze the mean squared interparticle separation $\langle \delta \mathbf{R}^2(t) \rangle$. The combined influence of elastic coupling and active driving force is governed by the response function $\Phi(t)$, which is symmetric under exchange of passive and active particles. Consequently, the two parameter regimes $\alpha_A > \alpha_P$ and $\alpha_A < \alpha_P$ lead to qualitatively similar dynamical behavior. To better understand the physical implications of the model, we first examine several limiting cases. In the absence of coupling $k = 0$, $\Phi(t) = 1$. In this limit, the mean squared interparticle distance becomes simply the sum of the individual MSDs of the uncoupled active and passive particles:

$$\langle \delta \mathbf{R}^2(t) \rangle_{k=0} = \langle \delta \mathbf{r}_A^2(t) \rangle_{k=0} + \langle \delta \mathbf{r}_P^2(t) \rangle_{k=0} \quad (40)$$

In the limit of very strong coupling $k \rightarrow \infty$, the response function approaches zero $\Phi(t) \rightarrow 0$ and the interparticle separation becomes strongly constrained by the elastic interaction. As a result, the mean squared interparticle distance saturates at the thermal equilibrium value

$$\langle \delta \mathbf{R}^2(t) \rangle_{k \rightarrow \infty} = \frac{4k_B T}{k} \quad (41)$$

Figure 5 illustrates the time evolution of $\langle \delta \mathbf{R}^2(t) \rangle$ for both weak and strong coupling regimes. In short-time, the influence of the active propulsion is negligible, the dynamics are dominated by thermal fluctuations and the

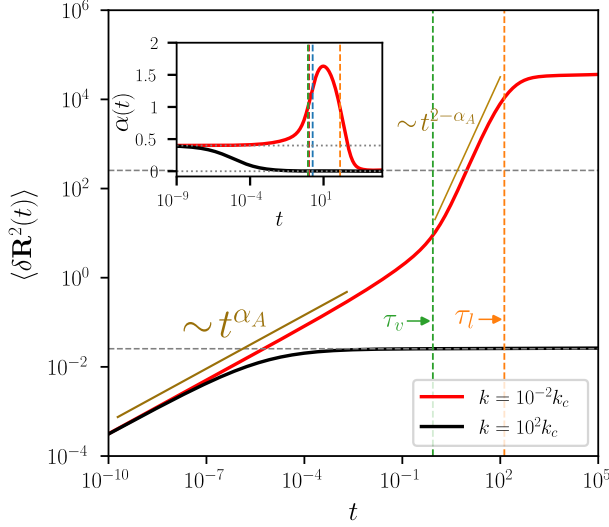


FIG. 5. Mean squared inter particle distance $\langle \delta \mathbf{R}^2(t) \rangle$ for weak and strong coupling regime. The parameters used in the figure are: $k_B T = 1, v = 2, \alpha_A = 0.4, \alpha_P = 0.8, \alpha_R = 0.5, \gamma_A = 1, \gamma_P = 1, \gamma_R = 4$. Inset shows $\alpha(t)$ for corresponding color.

interparticle separation follows

$$\langle \delta \mathbf{R}^2(t) \rangle \simeq \frac{4k_B T}{\eta \Gamma(1 + \alpha_{\min})} t^{\alpha_{\min}}. \quad (42)$$

where $\alpha_{\min} = \min(\alpha_A, \alpha_P)$ is the smaller of the two viscoelastic exponents, and η is the corresponding generalized friction coefficient, i.e., $\eta = \eta_A$ if $\alpha_{\min} = \alpha_A$, and $\eta = \eta_P$ if $\alpha_{\min} = \alpha_P$. Physically, at short times the interparticle separation is dominated by the particle with smaller exponent, since its viscoelastic response persists longest before the harmonic coupling becomes effective. For strong coupling, the system reaches a thermal plateau, indicating that the elastic interaction suppresses further growth of the interparticle separation. In the case of weak coupling, the active propulsion significantly enhances the interparticle fluctuations beyond the thermal plateau level. The system eventually approaches a steady state plateau at longer times due to the combined effects of elasticity and stochastic forcing.

C. Role of Anomalous Rotational Diffusion

Fig 6 shows the effect of α_R on the MSD of the active particle. The effect of α_R is only visible after the active force starts to influence after τ_v . For $\alpha_R = 0.5$ the MSD is higher compared to $\alpha_R = 1$. This happens because for small alpha the rotational orientation is highly correlated, therefore, the particle moves in the same direction. This results in a larger distance to travel.

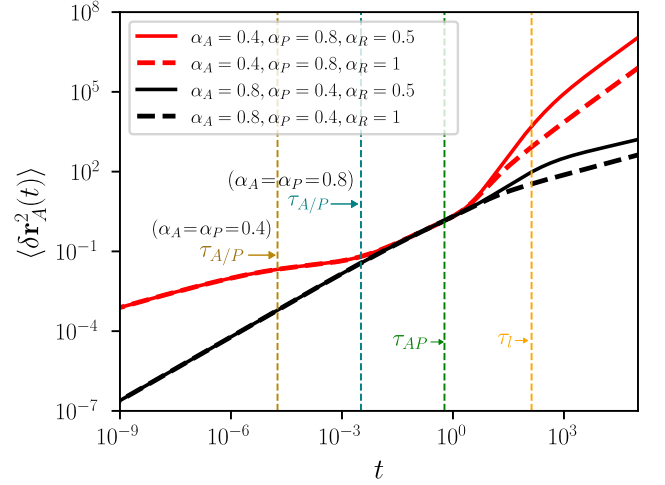


FIG. 6. MSD of active particle as a function of t , for different α_R . The parameters used in the figure are: $k_B T = 1, k = 100, v = 1, \gamma_A = 1, \gamma_P = 1, \gamma_R = 4$.

D. Non-Gaussian Parameter

Non-Gaussian diffusion is a feature of many complex systems, including crowded biological environments, soft glassy materials, and active matter systems. Many theoretical and experimental studies have reported deviations from Gaussian displacement statistics in systems ranging from intracellular transport to colloidal suspensions and active particles [116–121]. Such deviations often reflect underlying correlations, heterogeneity, or persistence in particle dynamics. To characterize these effects, we analyze both the self-part of the van Hove correlation function and the non-Gaussian parameter (NGP) using numerical simulations. The simulation methodology is given in Appendix F.

NGP provides a scalar measure of deviations from Gaussian statistics. We use time averaged moments of displacements to calculate this parameter, defined as [119, 122]

$$\alpha_2(\Delta t) = \frac{\overline{\Delta r^4(\Delta t)}}{2(\overline{\Delta r^2(\Delta t)})^2} - 1, \quad (43)$$

where

$$\overline{\Delta r^m(\Delta t)} = \frac{1}{T - \Delta t} \int_0^{T - \Delta t} |\mathbf{r}(t + \Delta t) - \mathbf{r}(t)|^m dt. \quad (44)$$

Here T is the full observation time of the trajectory. For Gaussian diffusion, we observe $\alpha_2(\Delta t) = 0$ while deviations from zero indicates non-Gaussianity. Positive values of NGP point towards broad distributions with long fat tails, whereas negative values correspond to narrower-than-Gaussian displacement distributions [119].

Since the present system is isotropic in two dimensions, we consider the displacement distribution along a single

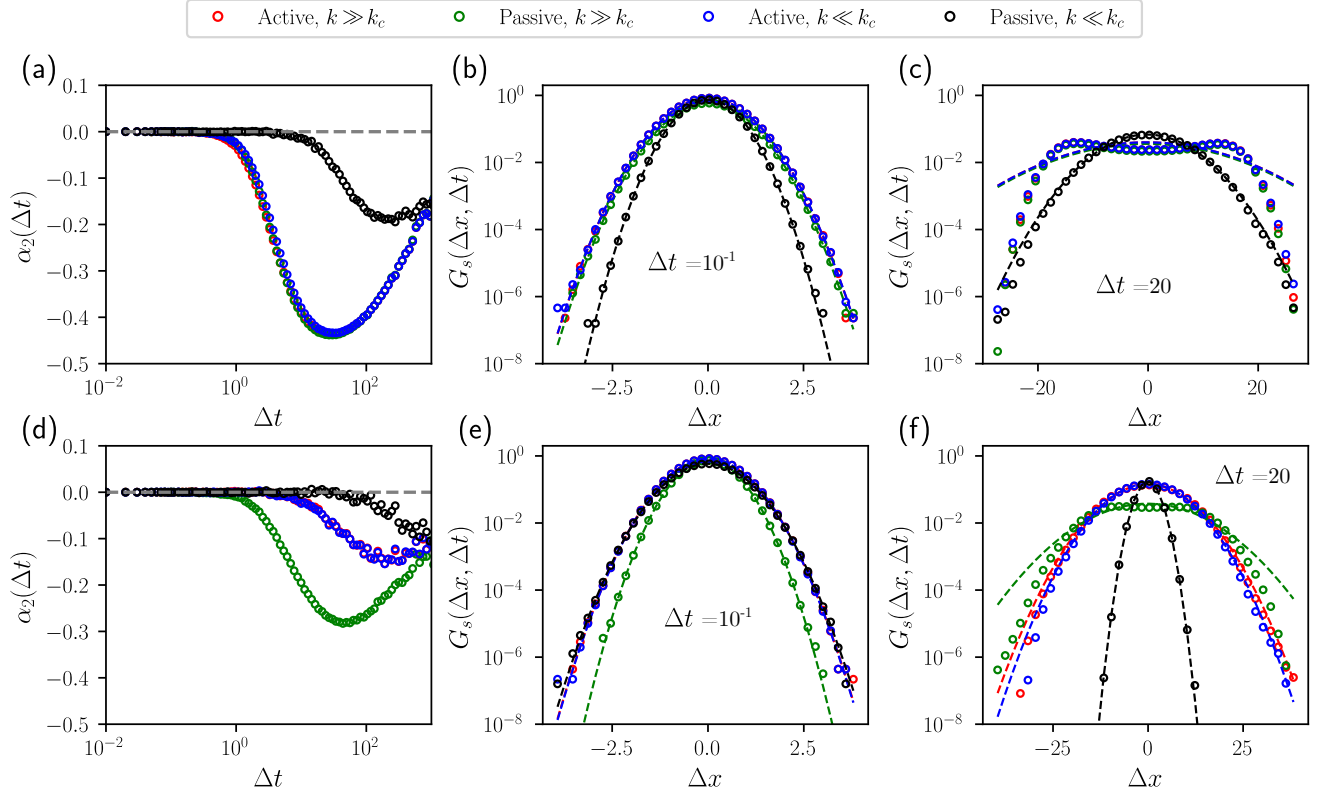


FIG. 7. Non-Gaussian parameter and the corresponding van Hove distributions for both active and passive particles under weak ($k \ll k_c$) and strong ($k \gg k_c$) coupling conditions. Panels (a)–(c) correspond to the case $\alpha_A < \alpha_P$, while panels (d)–(f) correspond to $\alpha_A > \alpha_P$. Panels (b) and (e) are for lag-time $\Delta t = 10^{-1}$; panels (c) and (f) are for lag-time $\Delta t = 20$. The circles represent simulation results, while the dotted curves in panels (b), (c), (e), and (f) denote Gaussian distributions with the same mean and variance as the corresponding simulation data. The parameters used in this figures are similar to Fig. 2, for corresponding cases.

x direction. The self-part of the time-averaged van Hove function is therefore defined as [123]

$$G_s(\Delta x, \Delta t) = \frac{1}{T - \Delta t} \int_0^{T - \Delta t} \delta(x(t + \Delta t) - x(t) - \Delta x) dt, \quad (45)$$

which gives the probability distribution of particle displacements Δx over a lag time Δt . For purely Gaussian diffusion, $G_s(\Delta x, \Delta t)$ assumes a Gaussian form whose width increases diffusively with time.

Since the bath noises $\xi_A(t)$, $\xi_P(t)$, $\xi_R(t)$ are Gaussian, and the orientation $\phi(t)$ is itself a Gaussian process, the fourth-order displacement moments entering the non-Gaussian parameter can in principle be computed exactly [124, 125]. For the present coupled active-passive system such expressions become algebraically cumbersome, and we therefore evaluate the non-Gaussian parameter via direct numerical simulation.

Figure 7a shows the non-Gaussian parameter $\alpha_2(\Delta t)$ as a function of lag time for the case of $\alpha_A < \alpha_P$. At short lag times $\alpha_2(\Delta t) \approx 0$, indicating that the displacement distribution is nearly Gaussian. In short lag times, the both particles do not get influenced by harmonic coupling

and active force. Here, particle motion is dominated by thermal stochastic fluctuations, and the dynamics closely resembles free subdiffusion. At intermediate lag times $\alpha_2(\Delta t)$ shows a negative minimum. This indicates a deviation from Gaussian statistics toward a sub-Gaussian regime. This behavior is reflected in the corresponding van Hove distributions. At short lag times, Fig. 7b, the distributions for both active and passive particles remain nearly Gaussian and are only weakly affected by the coupling strength. In contrast, at intermediate lag times, Fig. 7c, the distributions exhibit clear deviations from Gaussian behavior. The distributions show comparatively suppressed tails, consistent with the negative values of the non-Gaussian parameter. Here persistent motion with random orientation produces an effect of confinement. Similar sub-Gaussian behavior associated with persistence has been reported in active particle systems [126]. The strongest deviation in the Gaussianity of passive particle is observed for $k \gg k_c$ indicating that harmonic coupling enhances the transfer of temporal correlations between active and passive particles.

A qualitatively similar but quantitatively distinct behavior is observed for the case $\alpha_A > \alpha_P$, shown in Figs.

7d-f. In Fig. 7d, the non-Gaussian parameter again remains close to zero at short lag times before developing a negative minimum at intermediate times. Compared to the $\alpha_A < \alpha_P$ case, the magnitude of the deviation is reduced. In this case, the passive particle dynamics dominates the overall motion and reduces the influence of the active particle through harmonic coupling. The transfer of activity-induced correlations from the active particle to the passive particle becomes weaker, leading to comparatively smaller deviations from Gaussian behavior. The corresponding van Hove distributions shown in Figs. 7e and 7f further support this interpretation. These results show that the emergence and strength of non-Gaussian dynamics are strongly governed by the interplay between the viscoelastic properties of the medium and the coupling between active and passive particles.

IV. SUMMARY AND CONCLUSIONS

Understanding transport in complex environments is a central problem in soft matter and biological physics, where viscoelasticity, crowding, and activity often act together. In many realistic systems, particles are not isolated but interact through binding or mechanical coupling, which can significantly alter their motion. Motivated by such situations, we studied a minimal model of the active particle coupled to the passive particle through a harmonic interaction in a viscoelastic medium. We used the Generalized Langevin equation and numerical simulations based on fractional Brownian motion.

Our results demonstrate that the interplay between viscoelastic memory, elastic coupling, and active self-propulsion gives rise to a rich variety of dynamical regimes and crossover behaviors. Specifically, we analyzed the MSD for active and passive particles and mean squared interparticle separation. These quantities show that coupling introduces a nontrivial competition between activity and viscoelastic memory.

Depending on the relative subdiffusive exponents of the two particles, the long-time dynamics can be dominated by the active or passive component. In the case where the active particle is more subdiffusive ($\alpha_A < \alpha_P$), it can drag the passive particle and enhance transport that leads to regimes that include ballistic and normal diffusive behavior. In contrast, when the passive particle exhibits stronger subdiffusion ($\alpha_A > \alpha_P$), its sluggish dy-

namics suppress the contribution of active self-propulsion which shows long-time subdiffusive behavior even for the active particle.

We also identify crossovers between different dynamical regimes that are controlled by characteristic timescales associated with coupling, memory, and rotational diffusion. The interparticle separation reflects this interplay as well, showing confinement in the strong-coupling limit and enhanced fluctuations when the coupling is weak. We emphasize that these are genuinely emergent features of the coupling, not artifacts of the input exponents.

We also analyzed the non-Gaussian characteristics of dynamics using non-Gaussian parameter and van Hove displacement distributions. The system shows pronounced deviations from Gaussian behavior at intermediate lag times, originating from the combined influence of persistence, viscoelastic memory, and harmonic coupling. Stronger coupling enhances the transfer of temporal correlations between the particles and amplifies these non-Gaussian effects. Overall, the present study highlights how coupling between active and passive constituents in viscoelastic environments can fundamentally alter anomalous transport behavior.

Despite capturing several key features, the present model has some limitations. The present study is restricted to two dimensions; this is a limitation of the current work, and extending the framework to three dimensions is a natural direction for future work. The interaction between particles is modeled as a simple harmonic bond, which is permanent and does not allow for bond breaking or reformation, which is generally observed in real systems [22, 23, 127, 128]. Furthermore, active motion is described with a constant propulsion speed and simplified rotational dynamics with an isotropic environment, which may not capture more complex activity patterns seen in real systems [113, 129, 130]. These limitations point to several directions for future work. Finally, connecting theoretical predictions with experiments on tethered biomolecules, active gels, or synthetic microswimmers in complex media could further validate and refine the framework.

ACKNOWLEDGMENTS

PJ acknowledges IISER Pune for the Fellowship.

Appendix A: MSD Calculations

1. Position correlation function of the active particle

The position correlation function of the active particle in double Laplace space is

$$\begin{aligned} \langle \tilde{\mathbf{r}}_A(s) \tilde{\mathbf{r}}_A(s') \rangle &= \frac{\langle \mathbf{r}_A^2(0) \rangle}{ss'} + \frac{\tilde{\Psi}_P(s) \tilde{\Psi}_P(s') \langle \mathbf{R}^2(0) \rangle}{ss' (\tilde{K}_A(s) + \tilde{\Psi}_P(s)) (\tilde{K}_A(s') + \tilde{\Psi}_P(s'))} + v^2 \frac{\tilde{K}_A(s) \tilde{K}_A(s') \langle \hat{\mathbf{n}}(s) \hat{\mathbf{n}}(s') \rangle}{ss' (\tilde{K}_A(s) + \tilde{\Psi}_P(s)) (\tilde{K}_A(s') + \tilde{\Psi}_P(s'))} \\ &\quad + \frac{\frac{\tilde{\Psi}_P(s) \tilde{\Psi}_P(s')}{\tilde{K}_P(s) \tilde{K}_P(s')} \langle \tilde{\boldsymbol{\xi}}_P(s) \tilde{\boldsymbol{\xi}}_P(s') \rangle + \langle \tilde{\boldsymbol{\xi}}_A(s) \tilde{\boldsymbol{\xi}}_A(s') \rangle}{ss' (\tilde{K}_A(s) + \tilde{\Psi}_P(s)) (\tilde{K}_A(s') + \tilde{\Psi}_P(s'))}, \end{aligned} \quad (\text{A1})$$

where we have assumed the thermal noises acting on the active and passive particles are statistically independent,

$$\langle \boldsymbol{\xi}_A(t) \boldsymbol{\xi}_P(t') \rangle = \mathbf{0}. \quad (\text{A2})$$

a. Simplification of the noise term

We first simplify the last term on the right-hand side of Eq. (A1), which we denote L . Using the fluctuation-dissipation relations for $\tilde{\boldsymbol{\xi}}_A$ and $\tilde{\boldsymbol{\xi}}_P$, in Laplace space

$$\langle \tilde{\boldsymbol{\xi}}_A(s) \tilde{\boldsymbol{\xi}}_A(s') \rangle = 2k_B T \frac{\tilde{K}_A(s) + \tilde{K}_A(s')}{s + s'}, \quad \langle \tilde{\boldsymbol{\xi}}_P(s) \tilde{\boldsymbol{\xi}}_P(s') \rangle = 2k_B T \frac{\tilde{K}_P(s) + \tilde{K}_P(s')}{s + s'}. \quad (\text{A3})$$

Then, we can write

$$L = 2k_B T \frac{\tilde{\Psi}_P(s) \tilde{\Psi}_P(s') \left(\frac{1}{\tilde{K}_P(s)} + \frac{1}{\tilde{K}_P(s')} \right) + (\tilde{K}_A(s) + \tilde{K}_A(s'))}{ss' (s + s') (\tilde{K}_A(s) + \tilde{\Psi}_P(s)) (\tilde{K}_A(s') + \tilde{\Psi}_P(s'))}. \quad (\text{A4})$$

Adding and subtracting $(s + s')/k$ inside the bracket,

$$L = 2k_B T \frac{\tilde{\Psi}_P(s) \tilde{\Psi}_P(s') \left(\frac{1}{\tilde{K}_P(s)} + \frac{s}{k} + \frac{1}{\tilde{K}_P(s')} + \frac{s'}{k} - \frac{s + s'}{k} \right) + (\tilde{K}_A(s) + \tilde{K}_A(s'))}{ss' (s + s') (\tilde{K}_A(s) + \tilde{\Psi}_P(s)) (\tilde{K}_A(s') + \tilde{\Psi}_P(s'))}. \quad (\text{A5})$$

Using the identity $1/\tilde{K}_P(s) + s/k = 1/\tilde{\Psi}_P(s)$ [from the definition of $\tilde{\Psi}_P(s)$, Eq. (10)], Eq. (A5) becomes

$$L = 2k_B T \frac{\tilde{\Psi}_P(s) \tilde{\Psi}_P(s') \left(\frac{1}{\tilde{\Psi}_P(s)} + \frac{1}{\tilde{\Psi}_P(s')} - \frac{s + s'}{k} \right) + (\tilde{K}_A(s) + \tilde{K}_A(s'))}{ss' (s + s') (\tilde{K}_A(s) + \tilde{\Psi}_P(s)) (\tilde{K}_A(s') + \tilde{\Psi}_P(s'))}. \quad (\text{A6})$$

Expanding and regrouping terms, Eq. (A6) simplifies to

$$L = \frac{2k_B T}{s + s'} \left(\frac{1}{ss' (\tilde{K}_A(s) + \tilde{\Psi}_P(s))} + \frac{1}{ss' (\tilde{K}_A(s') + \tilde{\Psi}_P(s'))} \right) - \frac{2k_B T}{k} \frac{\tilde{\Psi}_P(s) \tilde{\Psi}_P(s')}{ss' (\tilde{K}_A(s) + \tilde{\Psi}_P(s)) (\tilde{K}_A(s') + \tilde{\Psi}_P(s'))}. \quad (\text{A7})$$

b. Full correlation function and its simplification

Substituting Eq. (A7) back into Eq. (A1) gives

$$\begin{aligned}
\langle \tilde{\mathbf{r}}_A(s) \tilde{\mathbf{r}}_A(s') \rangle &= \frac{\langle \mathbf{r}_A^2(0) \rangle}{ss'} + \frac{\tilde{\Psi}_P(s) \tilde{\Psi}_P(s') \langle \mathbf{R}^2(0) \rangle}{ss' (\tilde{K}_A(s) + \tilde{\Psi}_P(s)) (\tilde{K}_A(s') + \tilde{\Psi}_P(s'))} + v^2 \frac{\tilde{K}_A(s) \tilde{K}_A(s')}{ss' (\tilde{K}_A(s) + \tilde{\Psi}_P(s)) (\tilde{K}_A(s') + \tilde{\Psi}_P(s'))} \langle \hat{\mathbf{n}}(s) \hat{\mathbf{n}}(s') \rangle \\
&+ \frac{2k_B T}{s+s'} \left(\frac{1}{ss' (\tilde{K}_A(s) + \tilde{\Psi}_P(s))} + \frac{1}{ss' (\tilde{K}_A(s') + \tilde{\Psi}_P(s'))} \right) - \frac{2k_B T}{k} \frac{\tilde{\Psi}_P(s) \tilde{\Psi}_P(s')}{ss' (\tilde{K}_A(s) + \tilde{\Psi}_P(s)) (\tilde{K}_A(s') + \tilde{\Psi}_P(s'))}. \quad (\text{A8})
\end{aligned}$$

Using the equilibrium initial separation $\langle \mathbf{R}^2(0) \rangle = 2k_B T/k$, second and fourth terms on the right-hand side of Eq. (A8) exactly cancel. Recalling the auxiliary functions defined in Eqs. (12)–(13), Eq. (A8) thus reduces to

$$\langle \tilde{\mathbf{r}}_A(s) \tilde{\mathbf{r}}_A(s') \rangle = \frac{\langle \mathbf{r}_A^2(0) \rangle}{ss'} + \frac{2k_B T}{s+s'} \left(\frac{\tilde{\Theta}_A(s)}{s'} + \frac{\tilde{\Theta}_A(s')}{s} \right) + v^2 \tilde{\Delta}_A(s) \tilde{\Delta}_A(s') \langle \hat{\mathbf{n}}(s) \cdot \hat{\mathbf{n}}(s') \rangle. \quad (\text{A9})$$

c. Inversion to the time domain

Taking the inverse double Laplace transform of Eq. (A9) gives

$$\langle \mathbf{r}_A(t) \mathbf{r}_A(\tau) \rangle = \langle \mathbf{r}_A^2(0) \rangle + 2k_B T \int_0^t dt' \int_0^\tau d\tau' \delta(t' - \tau') [\Theta_A(t') + \Theta_A(\tau')] + v^2 \int_0^t dt' \int_0^\tau d\tau' \Delta_A(t') \Delta_A(\tau') C(|(t-t') - (\tau-\tau')|). \quad (\text{A10})$$

We now evaluate Eq. (A10) explicitly, retaining a non-zero initial position $\mathbf{r}_A(0)$, and show that it correctly reproduces the MSD given in Eq. (14) regardless of this choice.

a. Equal-time correlation. Setting $\tau = t$ in Eq. (A10) and carrying out the delta-function integration,

$$\int_0^t dt' \int_0^t d\tau' \delta(t' - \tau') [\Theta_A(t') + \Theta_A(\tau')] = 2 \int_0^t dt' \Theta_A(t'), \quad (\text{A11})$$

gives

$$\begin{aligned}
\langle \mathbf{r}_A^2(t) \rangle &= \langle \mathbf{r}_A^2(0) \rangle + 4k_B T \int_0^t dt' \Theta_A(t') \\
&+ v^2 \int_0^t dt' \int_0^t d\tau' \Delta_A(t') \Delta_A(\tau') C(|t' - \tau'|). \quad (\text{A12})
\end{aligned}$$

b. Correlation with the initial position. Setting $\tau = 0$ in Eq. (A10), the τ' -integration range collapses to a single point ($0 \leq \tau' \leq 0$), so both the thermal and active integrals vanish identically, leaving

$$\langle \mathbf{r}_A(t) \mathbf{r}_A(0) \rangle = \langle \mathbf{r}_A^2(0) \rangle. \quad (\text{A13})$$

This states that the particle's position at time t is, on average, correlated with its initial position *only* through $\langle \mathbf{r}_A^2(0) \rangle$ itself, that is as expected, since the thermal noise $\xi_A(t')$ and the active orientation $\hat{\mathbf{n}}(t')$ entering the dynamics for $t' > 0$ are uncorrelated with the initial condition $\mathbf{r}_A(0)$.

c. Mean-squared displacement. By definition,

$$\begin{aligned}
\langle \delta \mathbf{r}_A^2(t) \rangle &= \langle (\mathbf{r}_A(t) - \mathbf{r}_A(0))^2 \rangle \\
&= \langle \mathbf{r}_A^2(t) \rangle + \langle \mathbf{r}_A^2(0) \rangle - 2\langle \mathbf{r}_A(t) \mathbf{r}_A(0) \rangle. \quad (\text{A14})
\end{aligned}$$

Substituting Eqs. (A12) and (A13) into Eq. (A14), and simplifying we obtain Eq. (14). This confirms explicitly that the MSD is independent of $\langle \mathbf{r}_A^2(0) \rangle$. The result for $\langle \delta \mathbf{r}_A^2(t) \rangle$ is exact for arbitrary, even random, initial position $\mathbf{r}_A(0)$, provided only that $\mathbf{r}_A(0)$ is statistically independent of the subsequent noise and active-orientation fluctuations.

The same procedure, starting from Eq. (9), applies to the passive particle. After simplification and using auxiliary functions $\Theta_P(t)$ and $\Delta_P(t)$ one obtains Eq. (15).

Appendix B: Mean-squared interparticle separation

Rearranging Eq. (16) for $\tilde{\mathbf{R}}(s)$ gives

$$\begin{aligned}
\tilde{\mathbf{R}}(s) &= \frac{\tilde{K}_{eff}(s)}{s\tilde{K}_{eff}(s) + k} \mathbf{R}(0) + v \tilde{\mathbf{n}}(s) \frac{\tilde{K}_{eff}(s)}{s\tilde{K}_{eff}(s) + k} \\
&+ \frac{\tilde{\xi}_{eff}(s)}{s\tilde{K}_{eff}(s) + k}. \quad (\text{B1})
\end{aligned}$$

Introducing the auxiliary function $\tilde{\Phi}(s)$, defined in Eq. (22), Eq. (B1) simplifies to

$$\tilde{\mathbf{R}}(s) = \mathbf{R}(0) \tilde{\Phi}(s) + v \tilde{\mathbf{n}}(s) \tilde{\Phi}(s) + \frac{\tilde{\Phi}(s)}{\tilde{K}_{eff}(s)} \tilde{\xi}_{eff}(s). \quad (\text{B2})$$

1. Position correlation function in Laplace space

The two-point correlation function of $\mathbf{R}$ in double Laplace space, s and s' , follows directly from Eq. (B2):

$$\begin{aligned} \langle \tilde{\mathbf{R}}(s) \tilde{\mathbf{R}}(s') \rangle = & \langle \mathbf{R}^2(0) \rangle \tilde{\Phi}(s) \tilde{\Phi}(s') + v^2 \langle \tilde{\mathbf{n}}(s) \cdot \tilde{\mathbf{n}}(s') \rangle \tilde{\Phi}(s) \tilde{\Phi}(s') \\ & + \frac{\tilde{\Phi}(s) \tilde{\Phi}(s')}{\tilde{K}_{eff}(s) \tilde{K}_{eff}(s')} \langle \tilde{\xi}_{eff}(s) \tilde{\xi}_{eff}(s') \rangle. \end{aligned} \quad (\text{B3})$$

Using the double-Laplace-space form of the fluctuation-dissipation theorem for the effective noise,

$$\langle \tilde{\xi}_{eff}(s) \tilde{\xi}_{eff}(s') \rangle = 2k_B T \frac{\tilde{K}_{eff}(s) + \tilde{K}_{eff}(s')}{s + s'}, \quad (\text{B4})$$

Eq. (B3) becomes

$$\langle \tilde{\mathbf{R}}(s) \tilde{\mathbf{R}}(s') \rangle = \langle \mathbf{R}^2(0) \rangle \tilde{\Phi}(s) \tilde{\Phi}(s') + v^2 \langle \tilde{\mathbf{n}}(s) \cdot \tilde{\mathbf{n}}(s') \rangle \tilde{\Phi}(s) \tilde{\Phi}(s') + 2k_B T \frac{\tilde{\Phi}(s) \tilde{\Phi}(s')}{s + s'} \left(\frac{1}{\tilde{K}_{eff}(s)} + \frac{1}{\tilde{K}_{eff}(s')} \right). \quad (\text{B5})$$

2. Simplification using the equilibrium initial condition

Using the identity $\tilde{K}_{eff}(s) = k\tilde{\Phi}(s)/[1 - s\tilde{\Phi}(s)]$, which follows directly from the definition of $\tilde{\Phi}(s)$, for both s and s' in Eq. (B5) gives

$$\begin{aligned} \langle \tilde{\mathbf{R}}(s) \tilde{\mathbf{R}}(s') \rangle = & \langle \mathbf{R}^2(0) \rangle \tilde{\Phi}(s) \tilde{\Phi}(s') \\ & + \frac{2k_B T}{k} \left(\frac{\tilde{\Phi}(s) + \tilde{\Phi}(s')}{s + s'} - \tilde{\Phi}(s) \tilde{\Phi}(s') \right) \\ & + v^2 \langle \tilde{\mathbf{n}}(s) \cdot \tilde{\mathbf{n}}(s') \rangle \tilde{\Phi}(s) \tilde{\Phi}(s'). \end{aligned} \quad (\text{B6})$$

Imposing the equilibrium initial separation, $\langle \mathbf{R}^2(0) \rangle = 2k_B T/k$, the first term in Eq. (B6) exactly cancels the $-\tilde{\Phi}(s)\tilde{\Phi}(s')$ term in the bracket, leaving

$$\begin{aligned} \langle \tilde{\mathbf{R}}(s) \tilde{\mathbf{R}}(s') \rangle = & \frac{2k_B T}{k} \frac{\tilde{\Phi}(s) + \tilde{\Phi}(s')}{s + s'} \\ & + v^2 \langle \tilde{\mathbf{n}}(s) \cdot \tilde{\mathbf{n}}(s') \rangle \tilde{\Phi}(s) \tilde{\Phi}(s'). \end{aligned} \quad (\text{B7})$$

3. Inversion to the time domain

Taking the inverse double Laplace transform of Eq. (B7) yields

$$\begin{aligned} \langle \mathbf{R}(t) \mathbf{R}(\tau) \rangle = & \frac{2k_B T}{k} \Phi(|t - \tau|) \\ & + v^2 \int_0^t dt' \int_0^\tau d\tau' C(|(t - t') - (\tau - \tau')|) \\ & \times \Phi(t') \Phi(\tau'). \end{aligned} \quad (\text{B8})$$

The mean-squared interparticle separation is defined as

$$\begin{aligned} \langle \delta \mathbf{R}^2(t) \rangle = & \langle (\mathbf{R}(t) - \mathbf{R}(0))^2 \rangle \\ = & \langle \mathbf{R}^2(t) \rangle + \langle \mathbf{R}^2(0) \rangle - 2\langle \mathbf{R}(t) \mathbf{R}(0) \rangle. \end{aligned} \quad (\text{B9})$$

Setting $\tau = t$ in Eq. (B8) gives the mean-squared value of $\mathbf{R}(t)$,

$$\langle \mathbf{R}^2(t) \rangle = \frac{2k_B T}{k} + v^2 \int_0^t dt' \int_0^t d\tau' C(|t' - \tau'|) \Phi(t') \Phi(\tau'), \quad (\text{B10})$$

using $\Phi(0) = 1$, while setting $\tau = 0$ gives

$$\langle \mathbf{R}(t) \mathbf{R}(0) \rangle = \frac{2k_B T}{k} \Phi(t), \quad (\text{B11})$$

since the double integral in Eq. (B8) vanishes identically when $\tau = 0$. Substituting Eqs. (B10) and (B11), together with $\langle \mathbf{R}^2(0) \rangle = 2k_B T/k$, into Eq. (B9) gives the final expression for the mean-squared interparticle separation given in Eq. (21).

Appendix C: Calculation of the Orientational Correlation Function

From Eq. (4b), the Laplace-space solution for the orientation angle is

$$\tilde{\phi}(s) = \frac{\phi(0)}{s} + \frac{\tilde{\xi}_R(s)}{s\tilde{K}_R(s)}. \quad (\text{C1})$$

Defining $\tilde{G}(s) \equiv 1/[s\tilde{K}_R(s)] = s^{-\alpha_R}/\eta_R$ and taking the inverse Laplace transform of Eq. (C1) using the convolution theorem, we obtain

$$\phi(t) = \phi(0) + \int_0^t G(t - t') \xi_R(t') dt', \quad (\text{C2})$$

where the inverse Laplace transform of $\tilde{G}(s)$ is

$$G(t) = \frac{t^{\alpha_R - 1}}{\eta_R \Gamma(\alpha_R)}. \quad (\text{C3})$$

1. Two-time correlation function

Using Eq. (C2), the two-time correlation function of ϕ is

$$\langle \phi(t)\phi(\tau) \rangle = \langle \phi^2(0) \rangle + \int_0^t \int_0^\tau G(t-t') G(\tau-\tau') \times \langle \xi_R(t')\xi_R(\tau') \rangle dt' d\tau'. \quad (\text{C4})$$

Substituting the noise correlation from the fluctuation-dissipation theorem, Eq. (2) of the main text, $\langle \xi_R(t')\xi_R(\tau') \rangle = k_B T K_R(|t' - \tau'|)$, gives

$$\langle \phi(t)\phi(\tau) \rangle = \langle \phi^2(0) \rangle + k_B T \int_0^t \int_0^\tau G(t-t') G(\tau-\tau') \times K_R(|t' - \tau'|) dt' d\tau'. \quad (\text{C5})$$

Substituting $G(t)$ from Eq. (C3) and the memory kernel $K_R(t)$ [Eq. (3) of the main text] into Eq. (C5) and carrying out the double integration yields

$$\langle \phi(t)\phi(\tau) \rangle = \langle \phi^2(0) \rangle + \frac{k_B T}{\eta_R \Gamma(1 + \alpha_R)} (t^{\alpha_R} + \tau^{\alpha_R} - |t - \tau|^{\alpha_R}). \quad (\text{C6})$$

2. Mean-squared angular displacement

Setting $t = \tau$ in Eq. (C6) gives the mean-squared value of $\phi(t)$,

$$\langle \phi^2(t) \rangle = \langle \phi^2(0) \rangle + \frac{2k_B T}{\eta_R \Gamma(1 + \alpha_R)} t^{\alpha_R}. \quad (\text{C7})$$

The mean-squared angular displacement between times t and τ follows from

$$\langle (\phi(t) - \phi(\tau))^2 \rangle = \langle \phi^2(t) \rangle + \langle \phi^2(\tau) \rangle - 2\langle \phi(t)\phi(\tau) \rangle. \quad (\text{C8})$$

Substituting Eqs. (C6) and (C7) into Eq. (C8), the terms in $\langle \phi^2(0) \rangle$ and $t^{\alpha_R}, \tau^{\alpha_R}$ cancel, leaving

$$\langle (\phi(t) - \phi(\tau))^2 \rangle = \frac{2k_B T}{\eta_R \Gamma(1 + \alpha_R)} |t - \tau|^{\alpha_R}. \quad (\text{C9})$$

Appendix D: Auxiliary functions

The Laplace-domain auxiliary functions are

$$\tilde{\Theta}_A(s) = \frac{1}{\frac{\eta_P k s^{\alpha_P}}{k + \eta_P s^{\alpha_P}} + \eta_A s^{\alpha_A}}, \quad (\text{D1})$$

$$\tilde{\Theta}_P(s) = \frac{1}{\frac{\eta_A k s^{\alpha_A}}{k + \eta_A s^{\alpha_A}} + \eta_P s^{\alpha_P}}, \quad (\text{D2})$$

$$\tilde{\Delta}_A(s) = \frac{1}{\frac{\eta_P k s^{1-\alpha_A+\alpha_P}}{\eta_A k + \eta_A \eta_P s^{\alpha_P}} + s}, \quad (\text{D3})$$

$$\tilde{\Delta}_P(s) = \frac{\eta_A k}{s[k(\eta_A + \eta_P s^{\alpha_P-\alpha_A}) + \eta_A \eta_P s^{\alpha_P}]}, \quad (\text{D4})$$

$$\tilde{\Phi}(s) = \frac{\eta_A \eta_P}{s[k\eta_A s^{-\alpha_P} + k\eta_P s^{-\alpha_A} + \eta_A \eta_P]}. \quad (\text{D5})$$

Closed-form inverse Laplace transforms of these functions can be obtained in the equal-order limit $\alpha_A = \alpha_P \equiv \alpha$. Throughout, we use the two-parameter Mittag-Leffler function $E_{\alpha,\nu}(z) = \sum_{n=0}^{\infty} z^n / \Gamma(\alpha n + \nu)$ [131] (so that $E_{\alpha,1} \equiv E_\alpha$), and define the shorthand

$$\zeta = \frac{k(\eta_A + \eta_P)}{\eta_A \eta_P}. \quad (\text{D6})$$

a. $\Theta_A(t)$ and $\Theta_P(t)$

For $\alpha_A = \alpha_P = \alpha$, rearrangement gives

$$\tilde{\Theta}_A(s) = \frac{s^{-\alpha}}{\eta_A + \eta_P} + \frac{\eta_P}{\eta_A(\eta_A + \eta_P)} (s^\alpha + \zeta)^{-1}, \quad (\text{D7})$$

whose term-by-term inversion yields

$$\Theta_A(t) = \frac{t^{\alpha-1}}{\eta_A + \eta_P} \left[\frac{1}{\Gamma(\alpha)} + \frac{\eta_P}{\eta_A} E_{\alpha,\alpha}(-\zeta t^\alpha) \right]. \quad (\text{D8})$$

By the same procedure (with $\eta_A \leftrightarrow \eta_P$),

$$\tilde{\Theta}_P(s) = \frac{s^{-\alpha}}{\eta_A + \eta_P} + \frac{\eta_A}{\eta_P(\eta_A + \eta_P)} (s^\alpha + \zeta)^{-1}, \quad (\text{D9})$$

so that

$$\Theta_P(t) = \frac{t^{\alpha-1}}{\eta_A + \eta_P} \left[\frac{1}{\Gamma(\alpha)} + \frac{\eta_A}{\eta_P} E_{\alpha,\alpha}(-\zeta t^\alpha) \right]. \quad (\text{D10})$$

b. $\Delta_A(t)$ and $\Delta_P(t)$

In the same limit $\alpha_A = \alpha_P = \alpha$, $\tilde{\Delta}_A(s)$ reduces to

$$\tilde{\Delta}_A(s) = \frac{1}{s} - \frac{\eta_P}{\eta_A + \eta_P} \left[\frac{1}{s} - \frac{s^{\alpha-1}}{s^\alpha + \zeta} \right], \quad (\text{D11})$$

which inverts to

$$\Delta_A(t) = 1 - \frac{\eta_P}{\eta_A + \eta_P} [1 - E_{\alpha,1}(-\zeta t^\alpha)] \quad (\text{D12})$$

Similarly, $\tilde{\Delta}_P(s)$ reduces to

$$\tilde{\Delta}_P(s) = \frac{\eta_A}{\eta_A + \eta_P} \left[\frac{1}{s} - \frac{s^{\alpha-1}}{s^\alpha + \zeta} \right], \quad (\text{D13})$$

giving

$$\Delta_P(t) = \frac{\eta_A}{\eta_A + \eta_P} [1 - E_{\alpha,1}(-\zeta t^\alpha)]. \quad (\text{D14})$$

c. $\Phi(t)$

In the limit $\alpha_A = \alpha_P \equiv \alpha$, $\tilde{\Phi}(s)$ reduces to

$$\tilde{\Phi}(s) = \frac{s^{\alpha-1}}{s^\alpha + \zeta}, \quad (\text{D15})$$

Using the one-parameter Mittag-Leffler pair we can write,

$$\Phi(t) = E_\alpha(-\zeta t^\alpha). \quad (\text{D16})$$

At $\alpha = 1$, $E_1(-\zeta t) = e^{-\zeta t}$, recovering ordinary exponential relaxation. For $\alpha_A \neq \alpha_P$, no single Mittag-Leffler function suffices and $\Phi(t)$ is instead given by the two-term fractional relaxation series [132]

$$\Phi(t) = \sum_{n=0}^{\infty} \sum_{m=0}^n (-1)^n \binom{n}{m} \left(\frac{k}{\eta_A} \right)^{n-m} \left(\frac{k}{\eta_P} \right)^m \frac{t^{\alpha_A(n-m) + \alpha_P m}}{\Gamma(1 + \alpha_A(n-m) + \alpha_P m)}, \quad (\text{D17})$$

which reduces to Eq. (D16) when $\alpha_A = \alpha_P$.

For $\alpha_A \neq \alpha_P$ the Laplace inversion is not trivial for Δ and Θ , therefore, we provide a list of long and short time limit in Table.II.

Appendix E: Reduced MSD

For $k = 0$, $\Theta_A(t) = t^{-1+\alpha_A}/(\eta_A \Gamma(\alpha_A))$ and $\Delta_A(t) = 1$, therefore, from Eq. (14) one can write

$$\begin{aligned} \langle \delta \mathbf{r}_A^2(t) \rangle &= \frac{4k_B T}{\eta_A \Gamma(1 + \alpha_A)} t^{\alpha_A} \\ &+ v^2 \int_0^t \int_0^t e^{-D_R |t' - \tau'|^{\alpha_R}} dt' d\tau'. \end{aligned} \quad (\text{E1})$$

Since the integral is symmetric in Eq. (E1), we can use the identity

$$\int_0^t \int_0^t f(|t' - \tau'|) dt' d\tau' = 2 \int_0^t (t-u) f(u) du, \quad (\text{E2})$$

and can write

$$\begin{aligned} \langle \delta \mathbf{r}_A^2(t) \rangle &= \frac{4k_B T}{\eta_A \Gamma(1 + \alpha_A)} t^{\alpha_A} + 2v^2 \left[t \int_0^t e^{-D_R u^{\alpha_R}} du \right. \\ &\quad \left. - \int_0^t u e^{-D_R u^{\alpha_R}} du \right] \end{aligned} \quad (\text{E3})$$

We now substitute $x = D_R u^{\alpha_R}$, therefore, $u = (x/D_R)^{1/\alpha_R}$. Then the first integral can be written using lower incomplete gamma function

$$\begin{aligned} \int_0^t e^{-D_R u^{\alpha_R}} du &= \frac{1}{\alpha_R} D_R^{-1/\alpha_R} \int_0^{D_R t^{\alpha_R}} x^{\frac{1}{\alpha_R}-1} e^{-x} dx \\ &= \frac{1}{\alpha_R} D_R^{-1/\alpha_R} \gamma\left(\frac{1}{\alpha_R}, D_R t^{\alpha_R}\right) \end{aligned} \quad (\text{E4})$$

Similarly second integral

$$\begin{aligned} \int_0^t u e^{-D_R u^{\alpha_R}} du &= \frac{1}{\alpha_R} D_R^{-2/\alpha_R} \int_0^{D_R t^{\alpha_R}} x^{\frac{2}{\alpha_R}-1} e^{-x} dx \\ &= \frac{1}{\alpha_R} D_R^{-2/\alpha_R} \gamma\left(\frac{2}{\alpha_R}, D_R t^{\alpha_R}\right) \end{aligned} \quad (\text{E5})$$

Putting everything together we obtain the final expression in Eq. (28).

Similarly for the passive particle in the reduced ($k = 0$) limit $\Theta_P(t) = t^{-1+\alpha_P}/(\eta_P \Gamma(\alpha_P))$ and $\Delta_P(t) = 0$ which directly gives reduced Eq. (30).

TABLE II. Short- and long-time asymptotic behavior for $\Theta_A(t)$, $\Theta_P(t)$, $\Delta_A(t)$, $\Delta_P(t)$, and $\Phi(t)$.

Function	$\alpha_A > \alpha_P$		$\alpha_P > \alpha_A$	
	$t \rightarrow 0$	$t \rightarrow \infty$	$t \rightarrow 0$	$t \rightarrow \infty$
$\Theta_A(t)$	$\frac{t^{\alpha_A-1}}{\eta_A \Gamma(\alpha_A)}$	$\frac{t^{\alpha_P-1}}{\eta_P \Gamma(\alpha_P)}$	$\frac{t^{\alpha_A-1}}{\eta_A \Gamma(\alpha_A)}$	$\frac{t^{\alpha_A-1}}{\eta_A \Gamma(\alpha_A)}$
$\Theta_P(t)$	$\frac{t^{\alpha_P-1}}{\eta_P \Gamma(\alpha_P)}$	$\frac{t^{\alpha_P-1}}{\eta_P \Gamma(\alpha_P)}$	$\frac{t^{\alpha_P-1}}{\eta_P \Gamma(\alpha_P)}$	$\frac{t^{\alpha_A-1}}{\eta_A \Gamma(\alpha_A)}$
$\Delta_A(t)$	1	$\frac{\eta_A}{\eta_P} \frac{t^{\alpha_P-\alpha_A}}{\Gamma(1+\alpha_P-\alpha_A)}$	1	1
$\Delta_P(t)$	$\frac{k}{\eta_P} \frac{t^{\alpha_P}}{\Gamma(1+\alpha_P)}$	$\frac{\eta_A}{\eta_P} \frac{t^{\alpha_P-\alpha_A}}{\Gamma(1+\alpha_P-\alpha_A)}$	$\frac{k}{\eta_P} \frac{t^{\alpha_P}}{\Gamma(1+\alpha_P)}$	1
$\Phi(t)$	1	$\frac{\eta_A}{k} \frac{t^{-\alpha_A}}{\Gamma(1-\alpha_A)}$	1	$\frac{\eta_P}{k} \frac{t^{-\alpha_P}}{\Gamma(1-\alpha_P)}$

Appendix F: Simulation Method

1. Fractional Langevin equations

We write Eq. (1) of the main text in terms of the Caputo fractional derivative,

$$\eta_A {}^C D_t^{\alpha_A} [\mathbf{r}_A(t) - v\mathbf{N}(t)] = \mathbf{F}_A(t) + \boldsymbol{\xi}_A(t), \quad (\text{F1a})$$

$$\eta_R {}^C D_t^{\alpha_R} \phi(t) = \xi_R(t), \quad (\text{F1b})$$

$$\eta_P {}^C D_t^{\alpha_P} \mathbf{r}_P(t) = \mathbf{F}_P(t) + \boldsymbol{\xi}_P(t), \quad (\text{F1c})$$

where $\mathbf{r}_A = (x_A, y_A)$ and $\mathbf{r}_P = (x_P, y_P)$ are the active and passive particle positions, respectively, $\phi(t)$ is the orientation angle of the active particle, and

$$\mathbf{N}(t) = \int_0^t (\cos \phi(\tau), \sin \phi(\tau)) d\tau. \quad (\text{F2})$$

The harmonic interparticle force is

$$\mathbf{F}_A(t) = -k[\mathbf{r}_A(t) - \mathbf{r}_P(t)], \quad \mathbf{F}_P(t) = -\mathbf{F}_A(t). \quad (\text{F3})$$

2. Volterra integral form

Using the definition of the Caputo derivative [133],

$${}^C D_t^\alpha f(t) = \frac{1}{\Gamma(1-\alpha)} \int_0^t \frac{\dot{f}(\tau)}{(t-\tau)^\alpha} d\tau, \quad (\text{F4})$$

Eq. (F1) can be rewritten in Volterra integral form [134] as

$$\mathbf{r}_A(t) = \mathbf{r}_A(0) + v\mathbf{N}(t) + \frac{1}{\eta_A \Gamma(\alpha_A)} \int_0^t (t-\tau)^{\alpha_A-1} \times [\mathbf{F}_A(\tau) + \boldsymbol{\xi}_A(\tau)] d\tau, \quad (\text{F5a})$$

$$\phi(t) = \phi(0) + \frac{1}{\eta_R \Gamma(\alpha_R)} \int_0^t (t-\tau)^{\alpha_R-1} \xi_R(\tau) d\tau, \quad (\text{F5b})$$

$$\mathbf{r}_P(t) = \mathbf{r}_P(0) + \frac{1}{\eta_P \Gamma(\alpha_P)} \int_0^t (t-\tau)^{\alpha_P-1} \times [\mathbf{F}_P(\tau) + \boldsymbol{\xi}_P(\tau)] d\tau. \quad (\text{F5c})$$

3. L1 discretization

We divide time into a uniform mesh $t_n = nh$, with timestep h . The Caputo derivative at t_n is approximated using the L1 scheme [135],

$${}^C D_t^\alpha f(t_n) \approx \frac{1}{\Gamma(2-\alpha) h^\alpha} \sum_{j=1}^n w(n-j) [f(t_j) - f(t_{j-1})], \quad (\text{F6})$$

where $w(l) = (l+1)^{1-\alpha} - l^{1-\alpha}$ are the L1 weights. For the translational and rotational degrees of freedom, these coefficients and the associated prefactors for $Q \in \{A, P, R\}$ are

$$w_Q(l) = (l+1)^{1-\alpha_Q} - l^{1-\alpha_Q} \quad (\text{F7})$$

and

$$\lambda_Q = \frac{\Gamma(2-\alpha_Q) h^{\alpha_Q}}{\eta_Q}. \quad (\text{F8})$$

4. Memory terms

The rotational equation is integrated as

$$\phi_n = \phi_{n-1} - \sum_{j=1}^{n-1} w_R(n-j) (\phi_j - \phi_{j-1}) + \lambda_R \xi_R^{(n)}. \quad (\text{F9})$$

Defining the angular increment $\Delta\phi_j = \phi_j - \phi_{j-1}$, the memory contribution to the rotational equation is

$$M_R^{(n)} = \sum_{j=1}^{n-1} w_R(n-j) \Delta\phi_j. \quad (\text{F10})$$

For sufficiently small h , the increment in the self-propulsion displacement is evaluated using the right-endpoint orientation,

$$\Delta\mathbf{N}^{(n)} = (h \cos \phi_n, h \sin \phi_n). \quad (\text{F11})$$

The corresponding memory contribution to the active particle's translational motion is

$$\mathbf{M}_A^{(n)} = \sum_{j=1}^{n-1} w_A(n-j) \mathbf{R}_A^{(j)}, \quad \mathbf{R}_A^{(j)} = v \Delta\mathbf{N}^{(j)} - \Delta\mathbf{r}_A^{(j)}. \quad (\text{F12})$$

For the passive particle, the self-propulsion term is absent, so

$$\mathbf{M}_P^{(n)} = \sum_{j=1}^{n-1} w_P(n-j) \mathbf{R}_P^{(j)}, \quad \mathbf{R}_P^{(j)} = -\Delta\mathbf{r}_P^{(j)}. \quad (\text{F13})$$

5. Predictor-corrector time integration

To improve the numerical stability of the coupled dynamics, the harmonic force is integrated using a

predictor-corrector scheme. The force evaluated at the previous timestep is

$$\mathbf{F}_A^{\text{old}} = -k(\mathbf{r}_A^{(n-1)} - \mathbf{r}_P^{(n-1)}), \quad \mathbf{F}_P^{\text{old}} = -\mathbf{F}_A^{\text{old}}. \quad (\text{F14})$$

Predictor step. Provisional positions are computed as

$$\bar{\mathbf{r}}_A^{(n)} = \mathbf{r}_A^{(n-1)} + v \Delta\mathbf{N}^{(n)} + \lambda_A \mathbf{F}_A^{\text{old}} + \mathbf{M}_A^{(n)} + \lambda_A \boldsymbol{\xi}_A^{(n)}, \quad (\text{F15})$$

$$\bar{\mathbf{r}}_P^{(n)} = \mathbf{r}_P^{(n-1)} + \lambda_P \mathbf{F}_P^{\text{old}} + \mathbf{M}_P^{(n)} + \lambda_P \boldsymbol{\xi}_P^{(n)}. \quad (\text{F16})$$

Predicted force. The force is then re-evaluated using the provisional positions,

$$\mathbf{F}_A^{\text{pred}} = -k(\bar{\mathbf{r}}_A^{(n)} - \bar{\mathbf{r}}_P^{(n)}), \quad \mathbf{F}_P^{\text{pred}} = -\mathbf{F}_A^{\text{pred}}. \quad (\text{F17})$$

Corrector step. The force used in the final update is the average of the old and predicted forces,

$$\mathbf{F}_Q^{\text{avg}} = \frac{1}{2}(\mathbf{F}_Q^{\text{old}} + \mathbf{F}_Q^{\text{pred}}), \quad Q \in \{A, P\}, \quad (\text{F18})$$

and the corrected positions are advanced as

$$\mathbf{r}_A^{(n)} = \mathbf{r}_A^{(n-1)} + v \Delta\mathbf{N}^{(n)} + \lambda_A \mathbf{F}_A^{\text{avg}} + \mathbf{M}_A^{(n)} + \lambda_A \boldsymbol{\xi}_A^{(n)}, \quad (\text{F19})$$

$$\mathbf{r}_P^{(n)} = \mathbf{r}_P^{(n-1)} + \lambda_P \mathbf{F}_P^{\text{avg}} + \mathbf{M}_P^{(n)} + \lambda_P \boldsymbol{\xi}_P^{(n)}. \quad (\text{F20})$$

The noise realizations $\boldsymbol{\xi}_A^{(n)}$ and $\boldsymbol{\xi}_P^{(n)}$ are held fixed between the predictor and corrector steps.

6. Noise generation

Fractional Gaussian noises were generated independently for each translational and rotational degree of freedom using Hosking's recursive method for fractional Gaussian noise synthesis [136]. The corresponding Hurst exponents satisfy

$$H_Q = 1 - \frac{\alpha_Q}{2}, \quad Q \in \{A, P, R\}, \quad (\text{F21})$$

consistent with the fluctuation-dissipation relation adopted in the theoretical model.

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